CEITEC

MUNI



C9555: A survival guide to simulations

Week 8

Denys Biriukov – Nov. 14th, 2024

Get ready for practical session

- ~~Open new terminal `Ctrl + Alt + T`~~

–

- ~~Course materials at:~~ https://gitlab.com/denysbiriukov/survival_guide_simulations_c9555

–

- ~~Any technical issues/problems/questions? Checklist:~~
  - ~~Access to Wolf.~~
  - ~~Proper infinity and conda environments.~~

- We will need only our browser today!

More a discussion than monologue

If you have a quick question to catch up, raise your hand and ask – we will try to fix it and bring you back

If quick answer does not help, let’s chat during a break or after

This course is heavily practical – theory will be trying to catch up, not vice versa

What is today?

- Week 8:
  Membranes in MD
- All cells have membrane(s)
- Membranes have lipids, proteins, saccharides, etc.
- What are lipids and lipid membranes/bilayers?
- How do we use MD to simulate lipids and membranes?

TEACHING WEEK	DATE	LECTURER	TOPIC
01	26.09	TR	Introduction and visualization with VMD
02	03.10	WM	MD Theory and Lennard-Jones Phases
03	10.10	WM	Lennard Jones Polymers
04	17.10	DB	How to run MD simulations?
05	24.10	DB	Adding salt to our water
06	31.10	TR	Pulling in MD
07	07.11	TR	Enhanced sampling in MD
08	14.11	DB	Membranes in MD
09	21.11	IK	Sampling in MD
10	28.11	IK	Free energies (1/2)
11	05.11	IK	Free energies (2/2)
12	12.12	WM	Using Alpha Fold in MD

Today program

Lecture

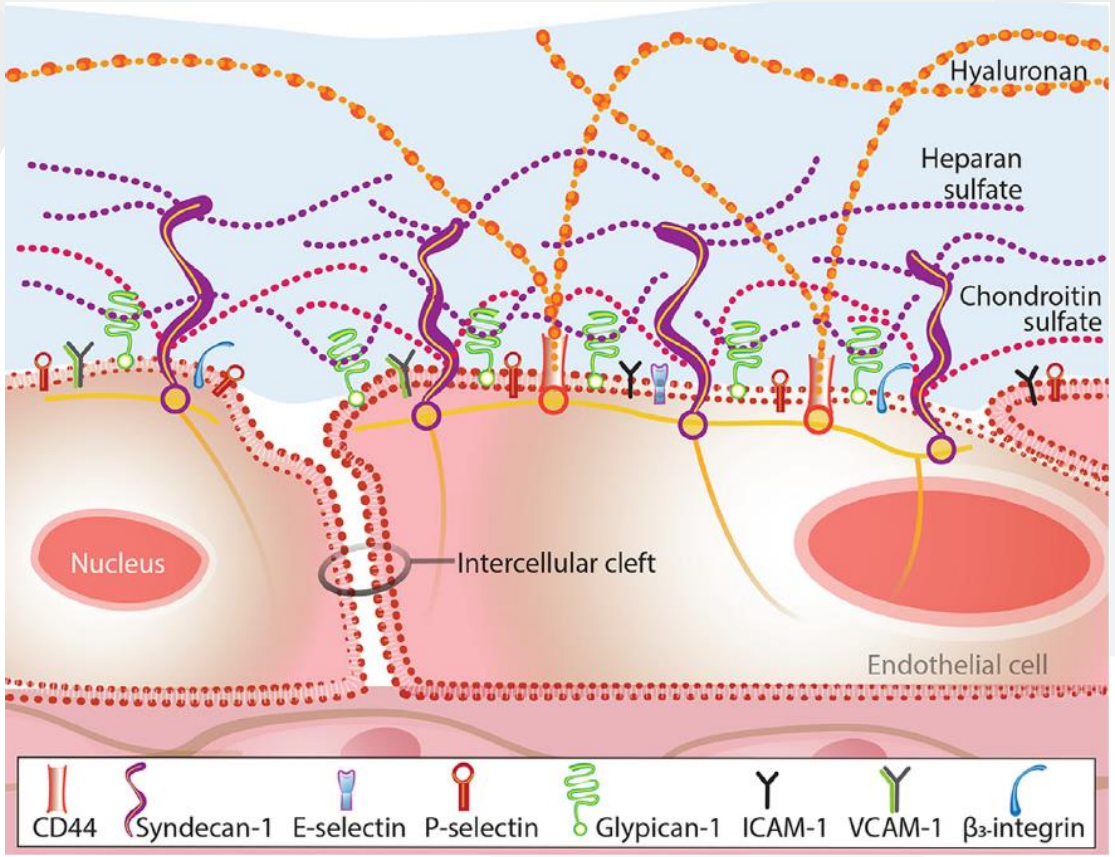
- Lipids in the context of cellular membranes.
- Structures, chemistry, and relevance of lipids.

Practical session

- Building a lipid bilayer using CHARMM-GUI online utility.
- Quick intro to analyzing MD simulations with membranes.
- Martini coarse-grained force field: lipids and not only.

Cellular membranes

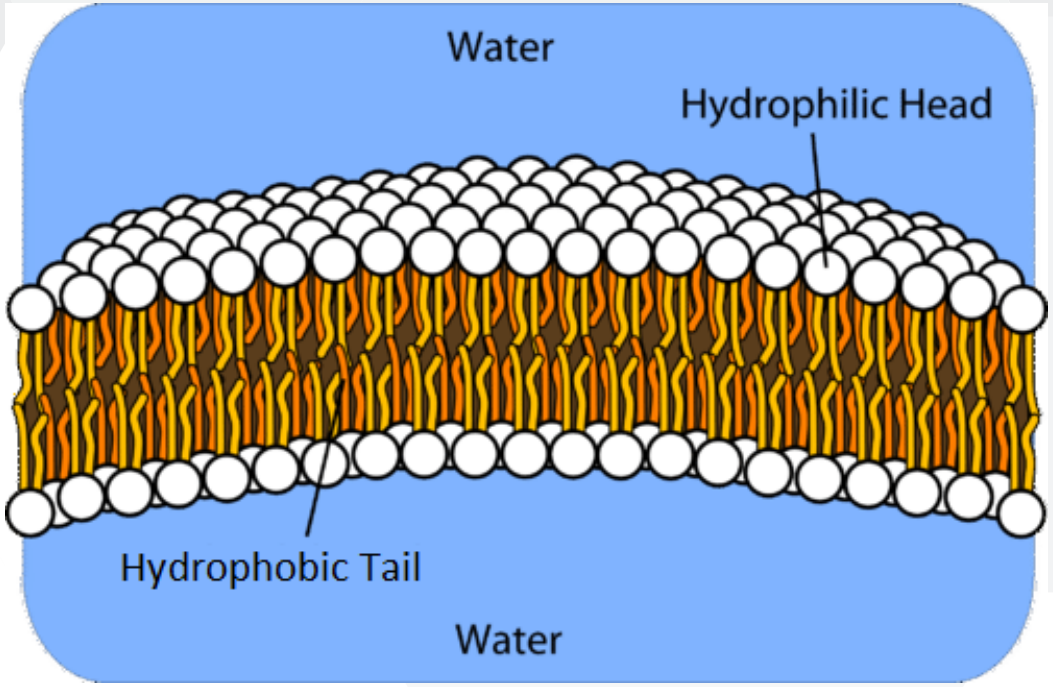
- Cells are the fundamental units of life.
- Every biological process, from metabolism to reproduction, occurs within cells.
- The cellular membrane is vital for maintaining the integrity of the cell.
- It provides a protective and selective barrier that separates the cell's internal environment from the external surroundings.



[Gaudette, S. et al. J Vet Emerg Crit Care. 2020, 1–18]

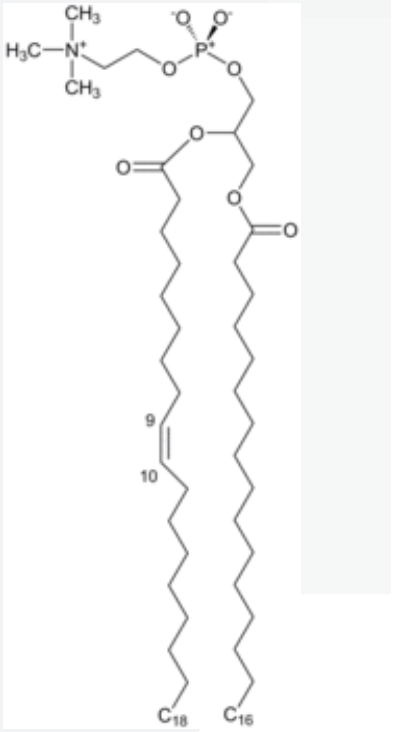
Lipid bilayers

- The membrane's structure is primarily composed of a phospholipid bilayer.
- Phospholipids (and also cholesterol) regulate the fluidity of the membrane, ensuring it remains functional and adaptable under various conditions.
- Lipids also create a suitable environment for membrane proteins to function effectively.
- Specific lipids can also organize and concentrate membrane proteins and signaling molecules, enhancing the efficiency and specificity of cellular signaling pathways.

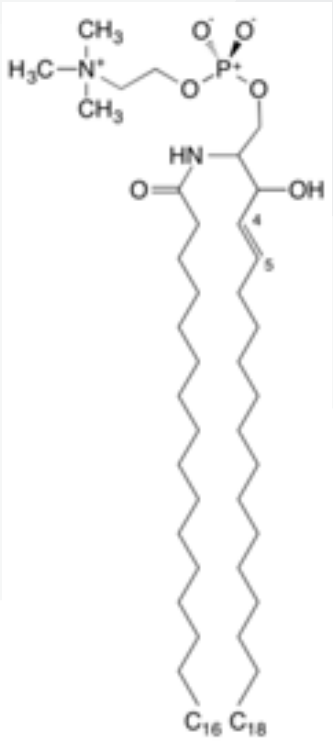


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Diversity of lipids

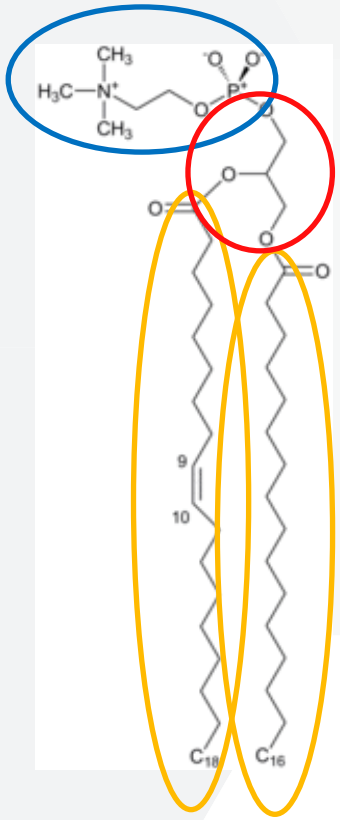


POPC



PSM

Diversity of lipids



POPC

headgroup (polar)

- **can be negatively charged**

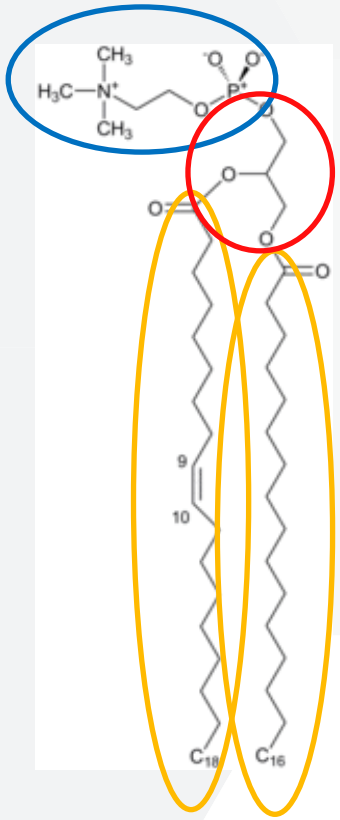
glycerol backbone

tails (fatty acids, nonpolar)

- **different length**
- **different saturation**

POPC – 1-palmitoyl-2-oleoyl-glycero-3-phosphocholine

Diversity of lipids

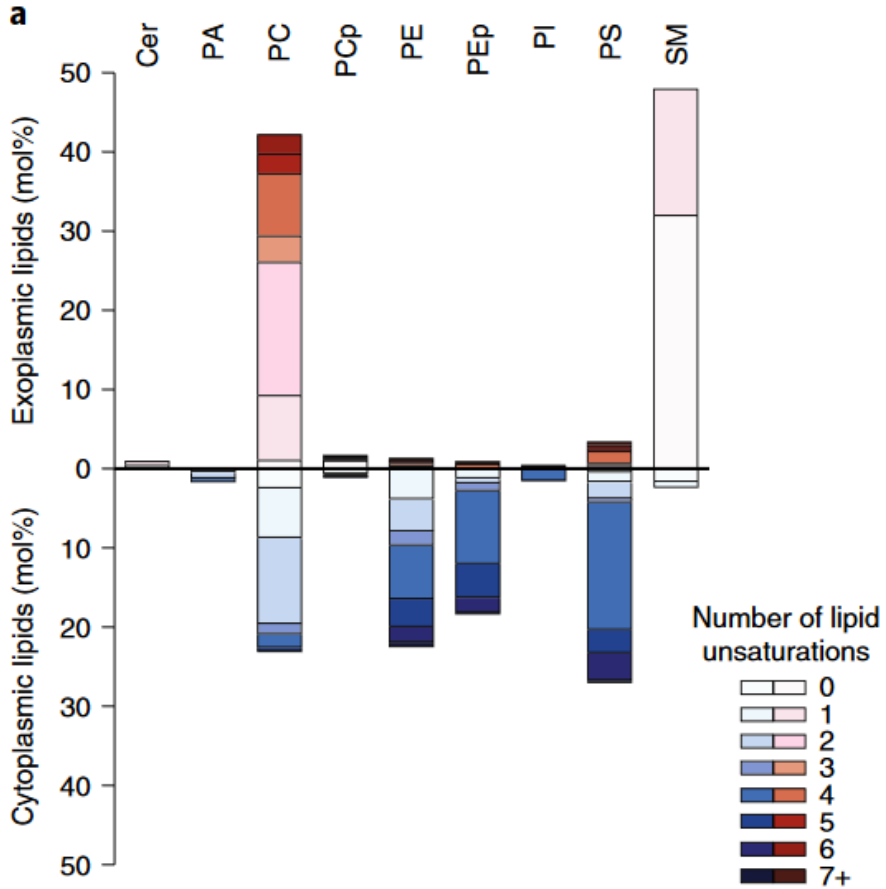


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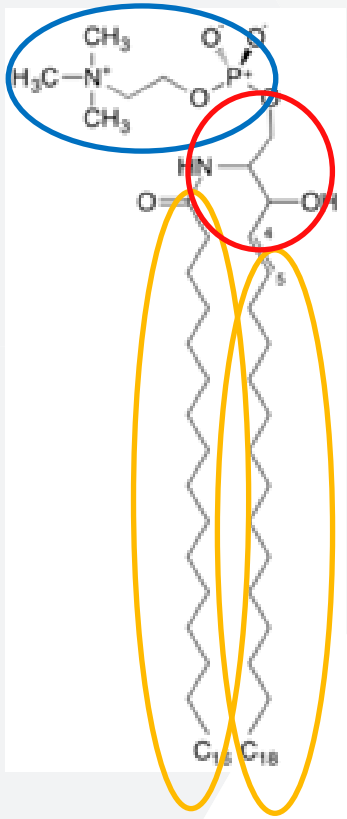


Lipids in plasma membrane of red blood cells

[Lorent, J. H. et al. Nat. Chem. Biol. 2020, 16, 644–652]

POPC – 1-palmitoyl-2-oleoyl-glycero-3-phosphocholine

Diversity of lipids



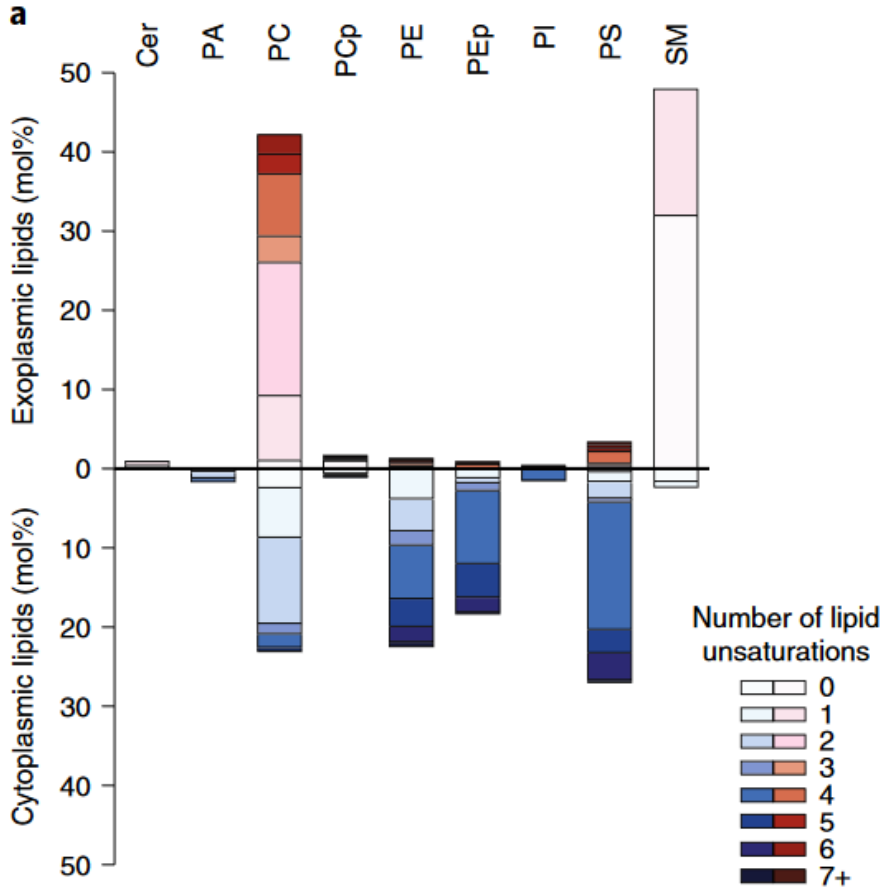
headgroup (polar)

sphingosine backbone

tails (fatty acids, nonpolar)

- **different length**
- **different saturation**

PSM

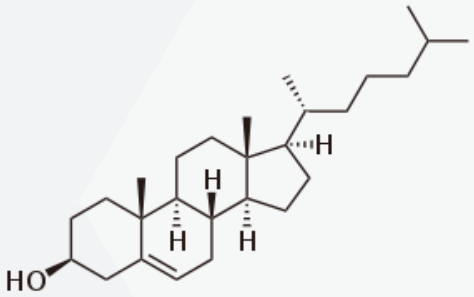


Lipids in plasma membrane of red blood cells

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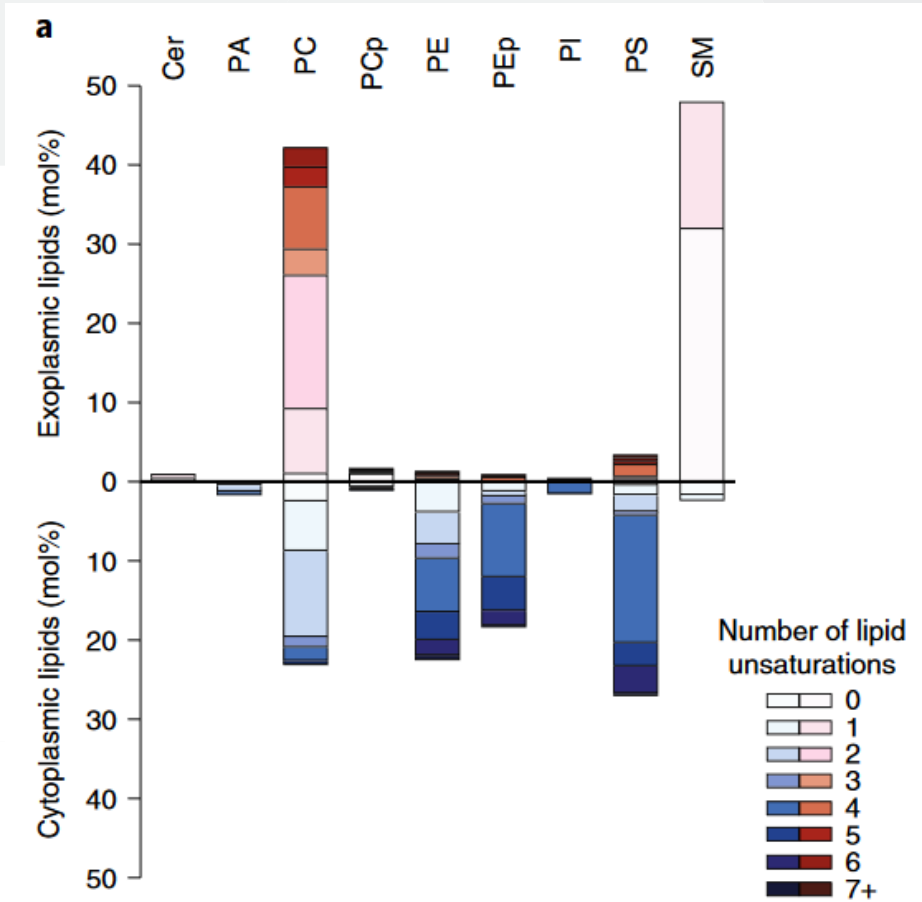
PSM – palmitoylsphingomyelin

Diversity of lipids



Cholesterol

- belongs to steroids
- amphiphilic
- mainly regulates fluidity

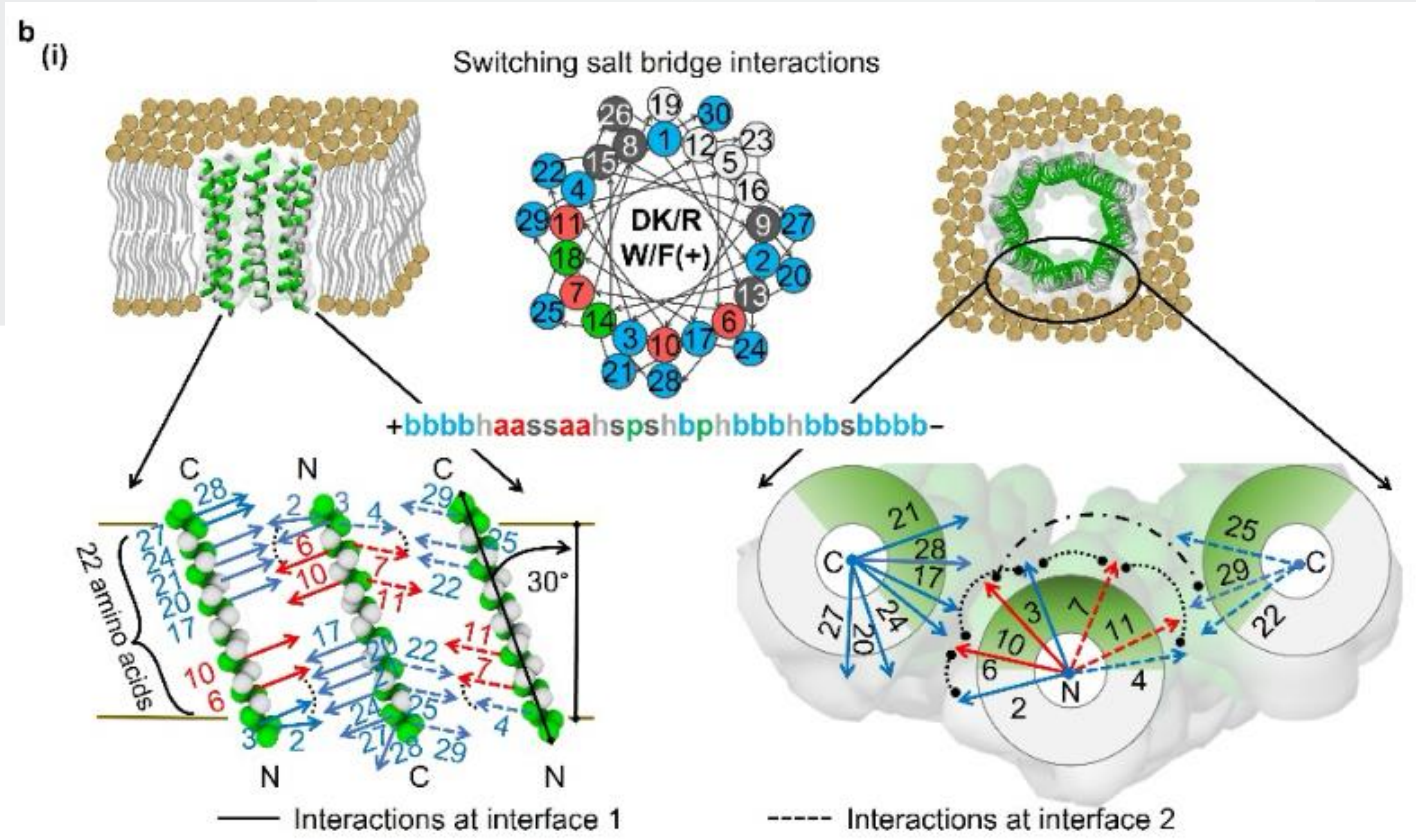


Lipids in plasma membrane of red blood cells

[Lorent, J. H. et al. Nat. Chem. Biol. 2020, 16, 644–652]

Why are lipids important?

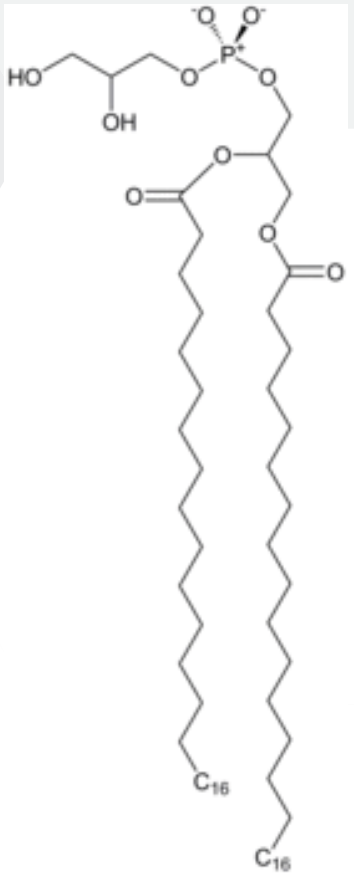
- All species possess membranes that contain lipids.
- Advanced simulations can be utilized to design novel drugs, such as antimicrobial peptides, tailored to target specific lipid compositions found in bacteria.
- These peptides can target and form pores in bacterial cell membranes, leading to bacterial cell death.



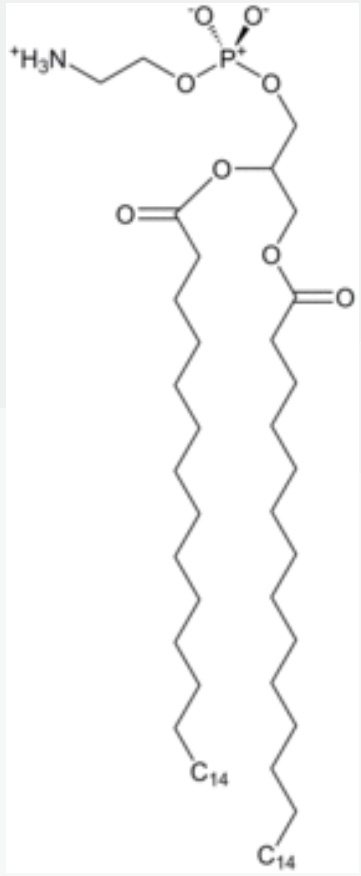
[Deb, R. et al. biorxiv, 2024, https://doi.org/10.1101/2022.05.09.491086]

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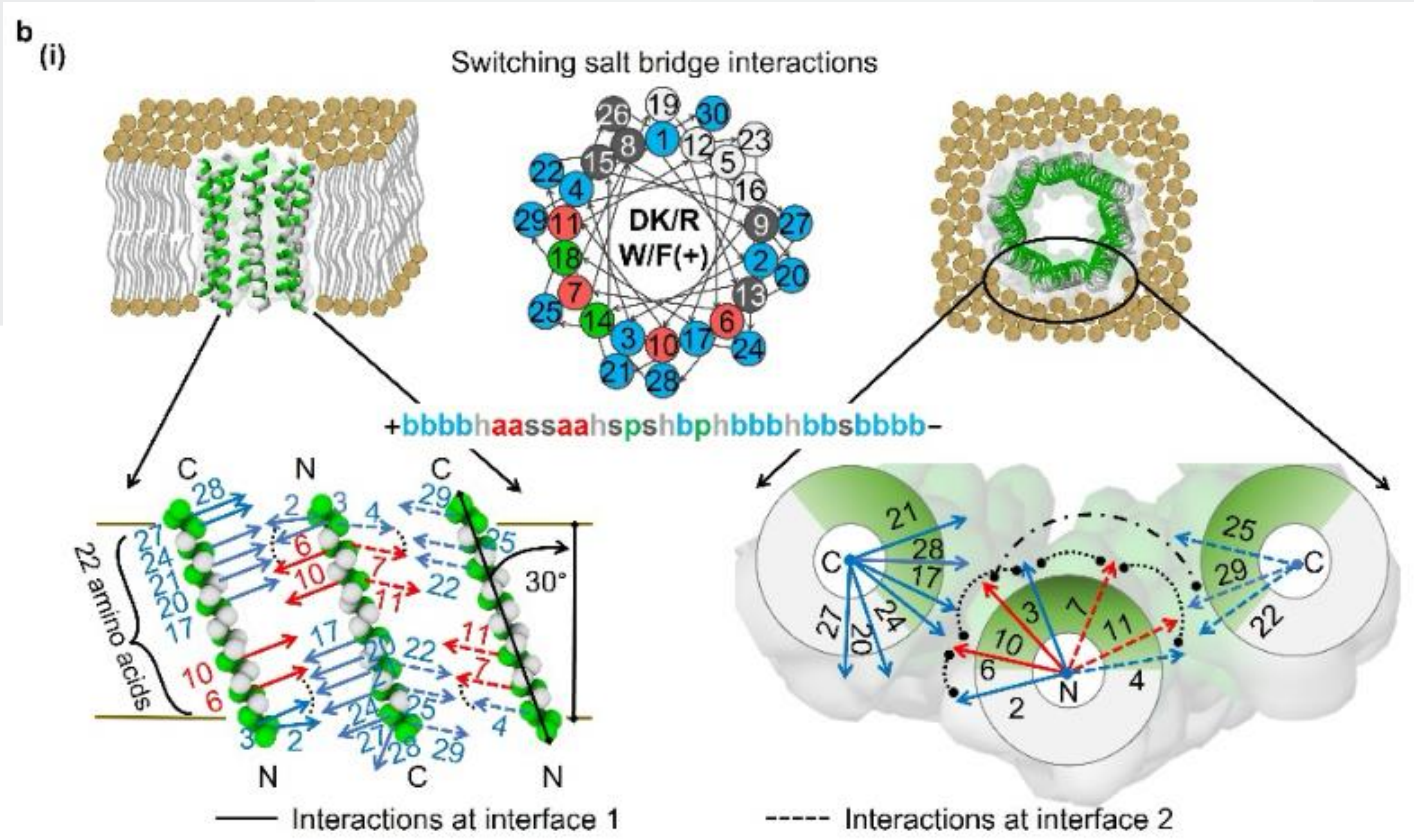
POPG



POPE

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Molecular dynamics of lipid membranes

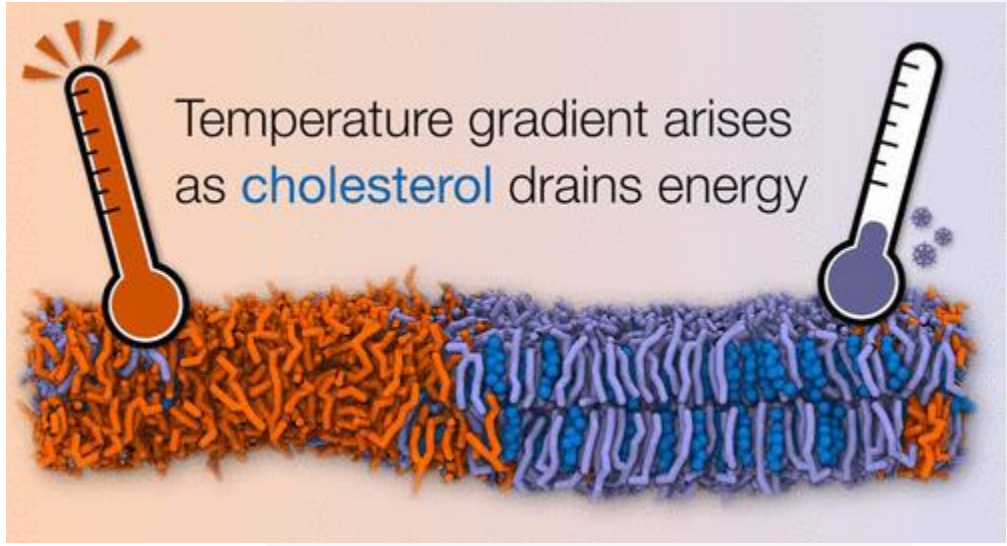
- I hope you know all theory behind MD by now. :)
- Solvated lipid bilayer is a heterogenous system.

Molecular dynamics of lipid membranes

- I hope you know all theory behind MD by now. :)
- Solvated lipid bilayer is a heterogenous system.
- Thermostatting a heterogeneous system (lipid bilayer in water, big protein in water, etc.) as a whole can create temperature gradients:
  - “Hot solvent – cold solute” problem.

Molecular dynamics of lipid membranes

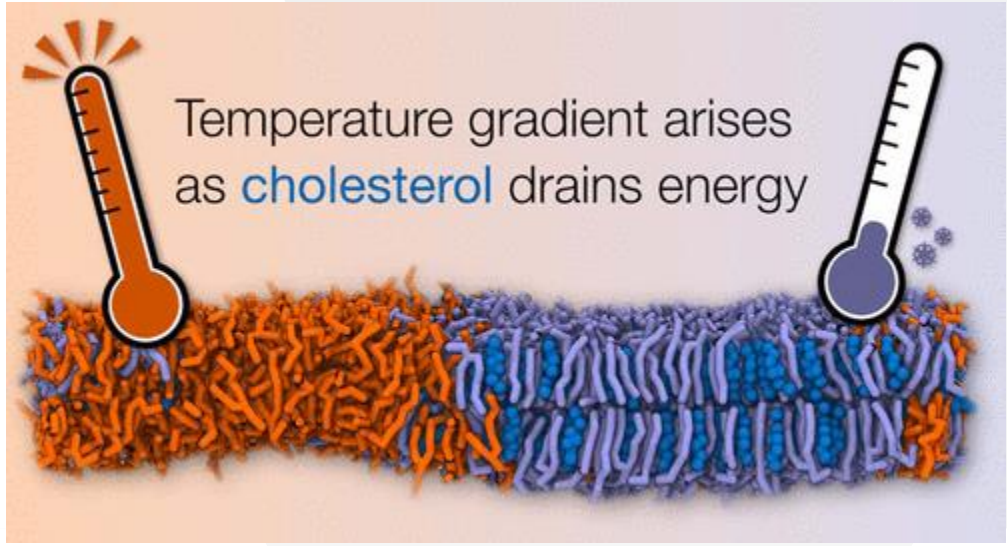
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  - "Hot solvent – cold solute" problem.
- Or even worse: improper simulation setup can lead to temperature gradients within the membrane.



[Thallmair et al. J. Phys. Chem. B 2021, 125, 33, 9537–9546]

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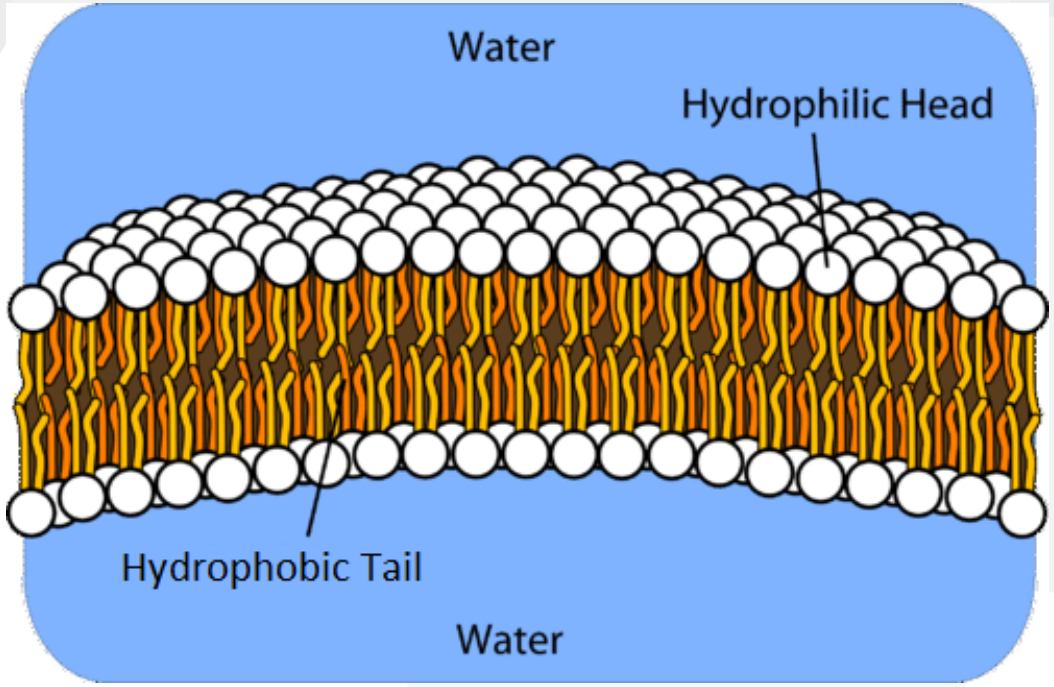


[Thallmair et al. J. Phys. Chem. B 2021, 125, 33, 9537–9546]

- **Solution:** thermostat membrane and solvent separately:
  - Decouples components with different dynamics
  - Prevents energy exchange artifacts

Molecular dynamics of lipid membranes

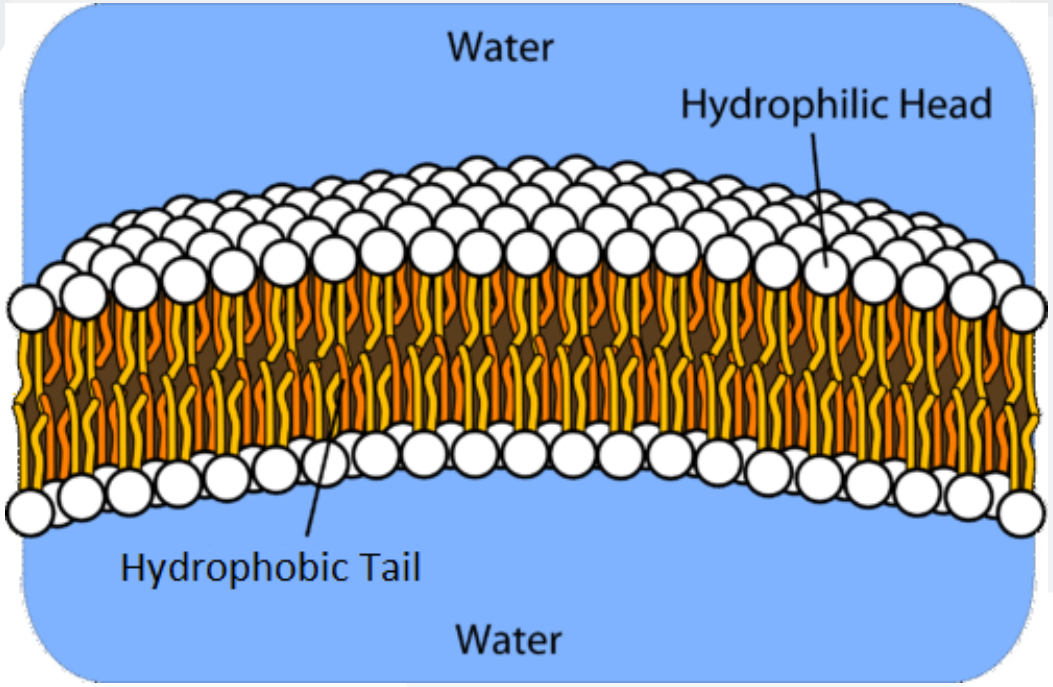
- The same logics applies to barostat.
- Lipid bilayer in water is an anisotropic environment.



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Molecular dynamics of lipid membranes

- The same logics applies to barostat.
- Lipid bilayer in water is an anisotropic environment.
- Semiisotropic barostat is required:
  - More accurate – structural anisotropy along and perpendicular to the membrane.
  - More realistic – cellular membranes experience different mechanical stress laterally and perpendicularly.
  - Prevents artifacts – deformations, bending, etc.



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Break?

Get ready for practical session!

CHARMM-GUI

CHARMM-GUI
Effective Simulation Input Generator and More

EXERCISE!

- CHARMM-GUI (https://www.charmm-gui.org/) is a web-based platform to interactively build complex, mainly biological systems
- Free, flexible, multiple submodules
- Provides input simulation files for multiple molecular dynamics engines (including GROMACS, Amber, CHARMM, …) and force fields (CHARMM, Amber, …)
- Today, we will build an all-atom model of a simple lipid bilayer mimicking plasma membrane

CHARMM-GUI

Effective Simulation Input Generator and More

CHARMM is a versatile program for atomic-level simulation of many-particle systems, particularly macromolecules of biological interest. - M. Karplus

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Some lectures, job postings, and FAQ are now available. See update log for update history and giving for donation. Contact info is given below.

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CHARMM-GUI

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Geographical Visitors



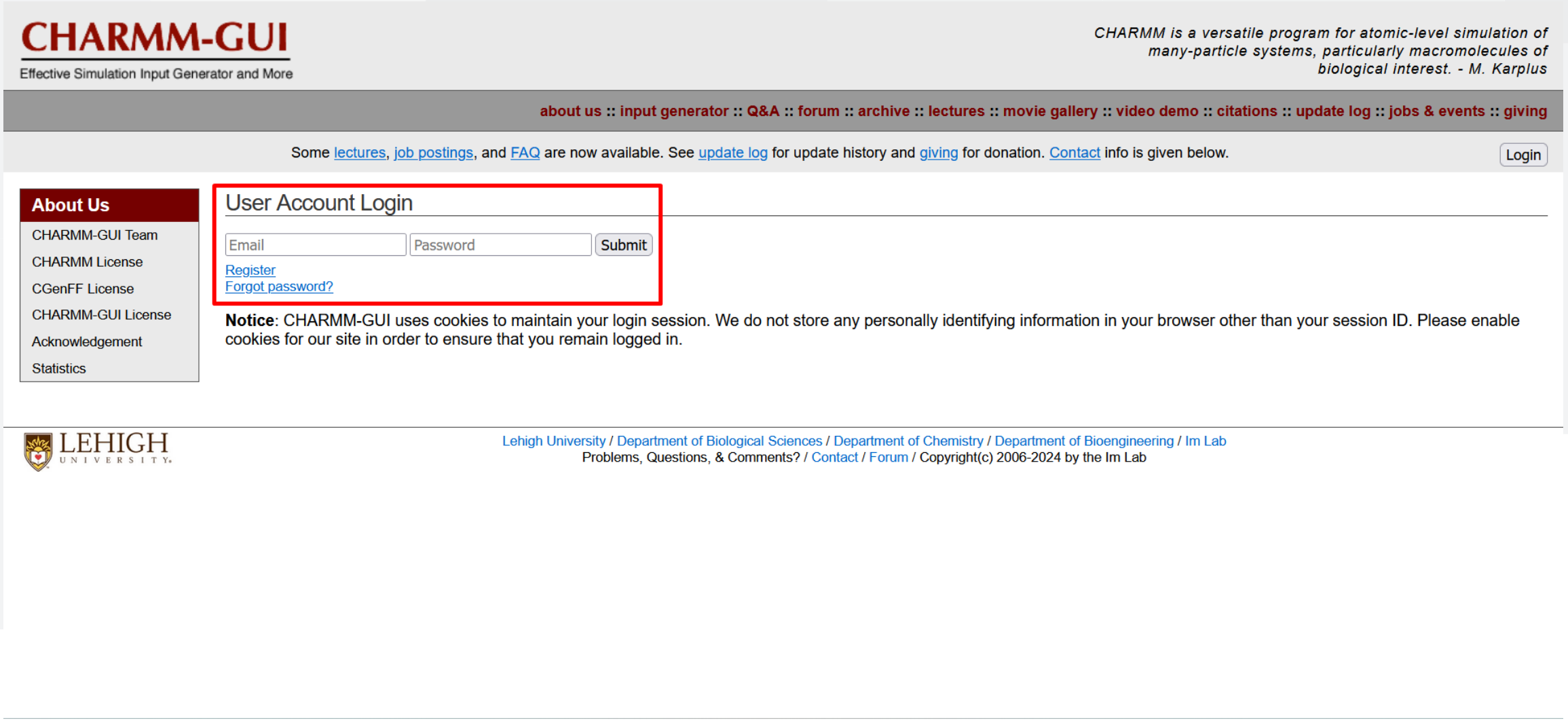
Front Page

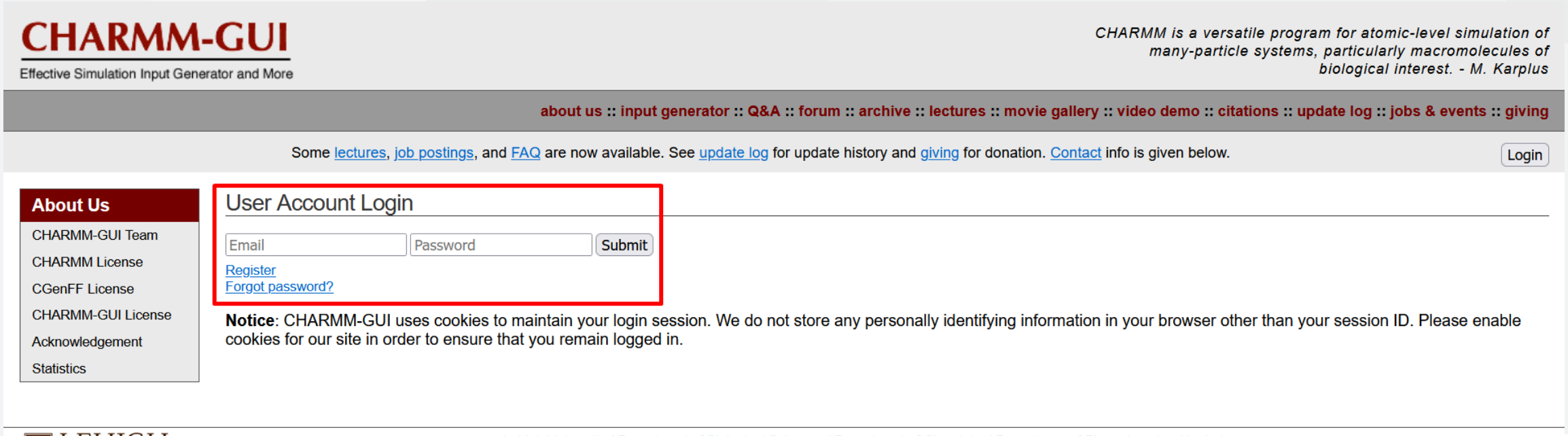
Since its original development in 2006, CHARMM-GUI has proven to be an ideal web-based platform to interactively build complex systems and prepare their inputs with well-established and reproducible simulation protocols for state-of-the-art molecular simulations using widely used simulation packages such as CHARMM, NAMD, GROMACS, AMBER, GENESIS, Tinker, LAMMPS, Desmond, and OpenMM. The CHARMM-GUI development project has been widely adopted for various purposes and now contains a number of different modules designed to set up a broad range of molecular simulation systems in Input Generator. Many original modules were developed as an in-house effort, but we have established close collaborations with the developers of CHARMM and other MD simulation packages for addition of newer modules.

Our philosophy in CHARMM-GUI development is less about providing the nuts and bolts of molecular modeling, but instead focused on helping users to achieve a task, such as building a membrane system or solvating a protein, by providing a streamlined interface. This design principle helps us to think of the workflow critically when designing the interface, which leads CHARMM-GUI to be accessible to users with little experience in modeling tools and remains useful to experts, especially for batch generation of systems. CHARMM-GUI has been used by many researchers, and it is a well-recognized tool in the molecular modeling and simulation communities (see Google Scholar Citations).

The CHARMM-GUI development project is still ongoing. These functionalities are not only based on requests from general users and developers, but also on an emerging need for a unified platform to prepare and execute various advanced simulation approaches that have been developed and will be developed by many developers in diverse simulation communities and packages. CHARMM-GUI will continue to help expert and non-expert researchers from a broader range of the modeling and simulation community to build the complex molecular systems of their interest and prepare the input files for any general and advanced modeling and simulation through the large and unique scope of CHARMM-GUI functionality. It will also provide an effective one-stop online resource for the biomedical research community to carry out innovative and novel molecular modeling and simulation research.

Visit our COVID-19 Archive for collection of SARS-CoV-2 protein systems.
Follow CHARMM-GUI on Twitter: https://twitter.com/CharmmGui.


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CHARMM is a versatile program for atomic-level simulation of many-particle systems, particularly macromolecules of biological interest. - M. Karplus
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Email: **denys.biriukov@ceitec.muni.cz**
Password: **studentC9555**

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User Profile

Input Generator

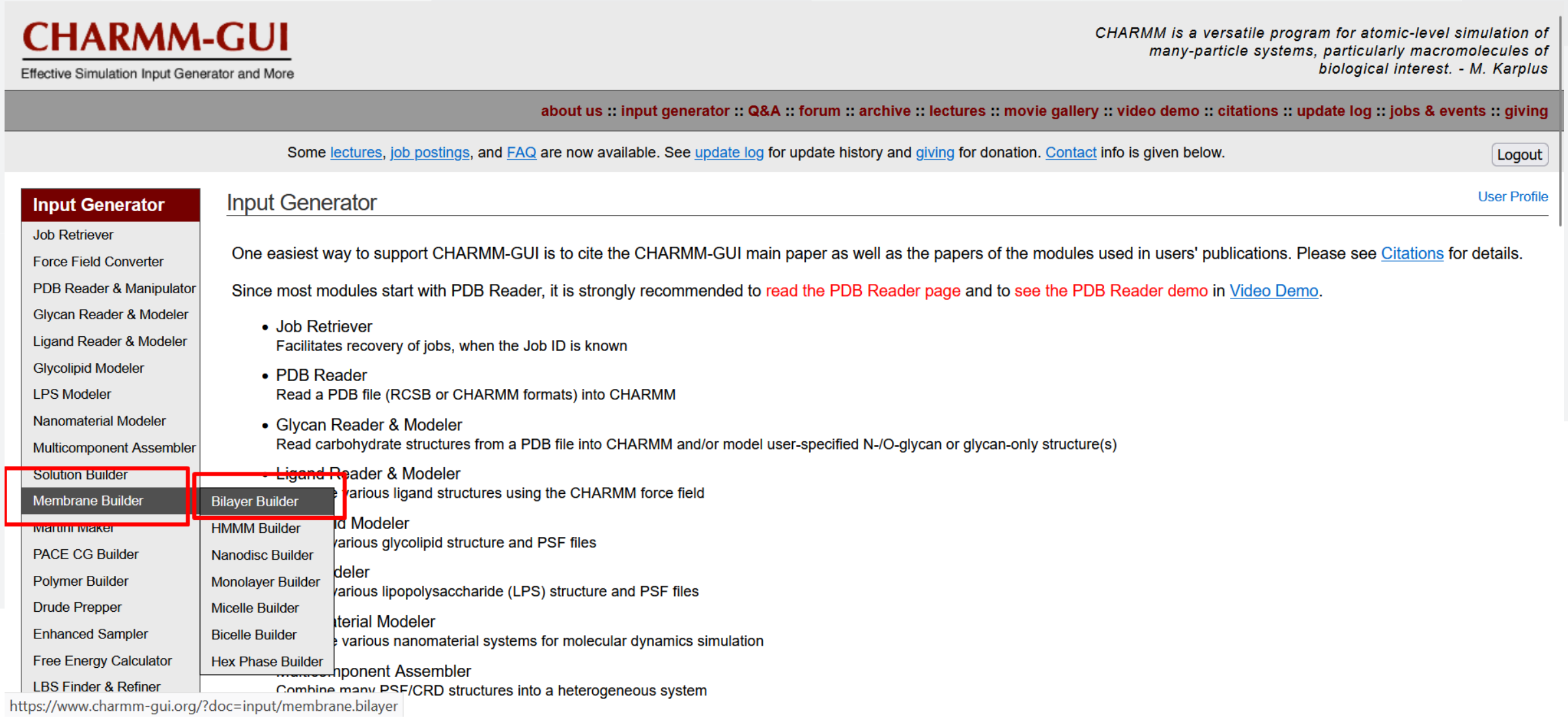
- Job Retriever
- Force Field Converter
- PDB Reader & Manipulator
- Glycan Reader & Modeler
- Ligand Reader & Modeler
- Glycolipid Modeler
- LPS Modeler
- Nanomaterial Modeler
- Multicomponent Assembler
- Solution Builder
- Membrane Builder
- Martini Maker
- PACE CG Builder
- Polymer Builder
- Drude Prepper
- Enhanced Sampler
- Free Energy Calculator
- LBS Finder & Refiner
- Ligand Designer

Input Generator

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Since most modules start with PDB Reader, it is strongly recommended to read the PDB Reader page and to see the PDB Reader demo in Video Demo.

- Job Retriever
  Facilitates recovery of jobs, when the Job ID is known
- PDB Reader
  Read a PDB file (RCSB or CHARMM formats) into CHARMM
- Glycan Reader & Modeler
  Read carbohydrate structures from a PDB file into CHARMM and/or model user-specified N-/O-glycan or glycan-only structure(s)
- Ligand Reader & Modeler
  Generate various ligand structures using the CHARMM force field
- Glycolipid Modeler
  Provide various glycolipid structure and PSF files
- LPS Modeler
  Provide various lipopolysaccharide (LPS) structure and PSF files
- Nanomaterial Modeler
  Generate various nanomaterial systems for molecular dynamics simulation
- Multicomponent Assembler
  Combine many PSF/CRD structures into a heterogeneous system


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https://www.charmm-gui.org/?doc=input/membrane.bilayer

CHARMM-GUI supports the Amber force fields. *J. Chem. Phys. 153:035103*

S. Park, Y.K. Choi, S. Kim, J. Lee, and W. Im (2021)
CHARMM-GUI Membrane Builder for Lipid Nanoparticles with Ionizable Cationic Lipids and PEGylated Lipids. *J. Chem. Inf. Model. 61:5192-5202*

○ **Protein/Membrane System**

Download PDB File: [ ] Download Source: [OPM]

Upload PDB File: [Browse...] No file selected.

PDB Format: ◉ PDB ○ PDBx/mmCIF ○ CHARMM

☐ Check/Correct PDB Format

◉ **Membrane Only System**

References for Lipid Force Fields:

J.B. Klauda, R.M. Venable, J.A. Freites, J.W. O'Connor, D.J. Tobias, C. Mondragon-Ramirez, I. Vorobyov, A.D. MacKerell, Jr., and R. W. Pastor (2010)
Update of the CHARMM all-atom additive force field for lipids: Validation on six lipid types *J Phys Chem B. 114(23): 7830–7843*

R.M. Venable, A.J. Sodt, B. Rogaski, H. Rui, E. Hatcher, A.D. MacKerell, Jr., R.W. Pastor, and J.B. Klauda. (2014)
CHARMM All-Atom Additive Force Field for Sphingomyelin: Elucidation of Hydrogen Bonding and of Positive Curvature. *Biophys J. 107(1):134-45*

Next Step: Select Lipids

A brief explanation of each step:

Today program

System Size Determination Options:

Homogeneous Lipid "Homogeneous Lipid" option is no longer supported. You can use "Heterogeneous Lipid" option even for homogeneous lipid bilayer building.

Heterogeneous Lipid

1. Box Type: Rectangular (Currently, only CHARMM, NAMD, and GROMACS support the hexagonal box)

2. Length of Z based on:
   - Water thickness 22.5 (Minimum water height on top and bottom of the system)
   - Hydration number 45 (Number of water molecules per one lipid molecule)

3. Length of XY based on:
   - Ratios of lipid components
   - Numbers of lipid components

XY Dimension Ratio: 1
(The system size along the X and Y must be the same)

Show the system info click this once you fill the following table:

- Both “water thickness” and “hydration number” are fine
- 45 water molecules is enough to model fully hydrated bilayers

Sterols						
Cholesterol	0		[Image]	30	30	40.0
PC (phosphatidylcholine) Lipids						
POPC	0	16:0 / 18:1	[Image]	35	35	68.3
SM (sphingo) and Ceramide Lipids						
PSM	0	18:1 / 16:0	[Image]	35	35	55.4

- Simple and quite classic mimic of the outer leaflet of the plasma membrane of red blood cells

XY Dimension Ratio: 1
(The system size along the X and Y must be the same)

Show the system info click this once you fill the following table:

Lipid Type	Charge [e]	Tail Info. [sn1/sn2]	Images	# of Lipid on Upperleaflet	# of Lipid on Lowerleaflet	Surface Area
▸ **Sterols**						
▸ **PA (phosphatidic acid) Lipids**						
▸ **PC (phosphatidylcholine) Lipids**						
▸ **PE (phosphatidylethanolamine) Lipids**						
▸ **PG (phosphatidylglycerol) Lipids**						
▸ **PS (phosphatidylserine) Lipids**						

Calculated XY System Size:

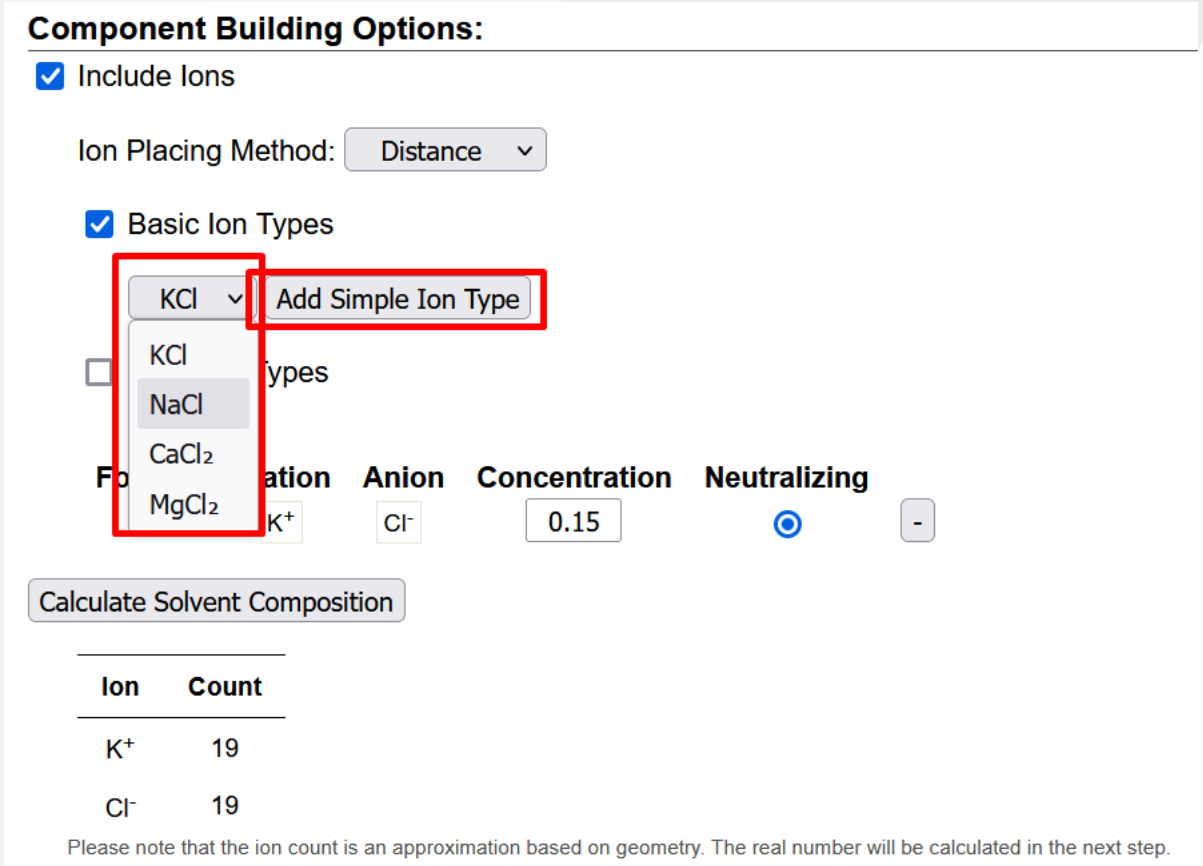
	Upperleaflet	Lowerleaflet
Protein Area	0	0
Lipid Area	5529.5	5529.5
# of Lipids	100	100
Total Area	5529.5	5529.5
Average Area	5529.50	
A	74.36	
B	74.36	

- **Detergents**
- **N-acylated Amino Acids**
- **PEG (polyethylene glycol) Lipids**
- **Glycolipids**
- **LPS (lipopolysaccharides)**

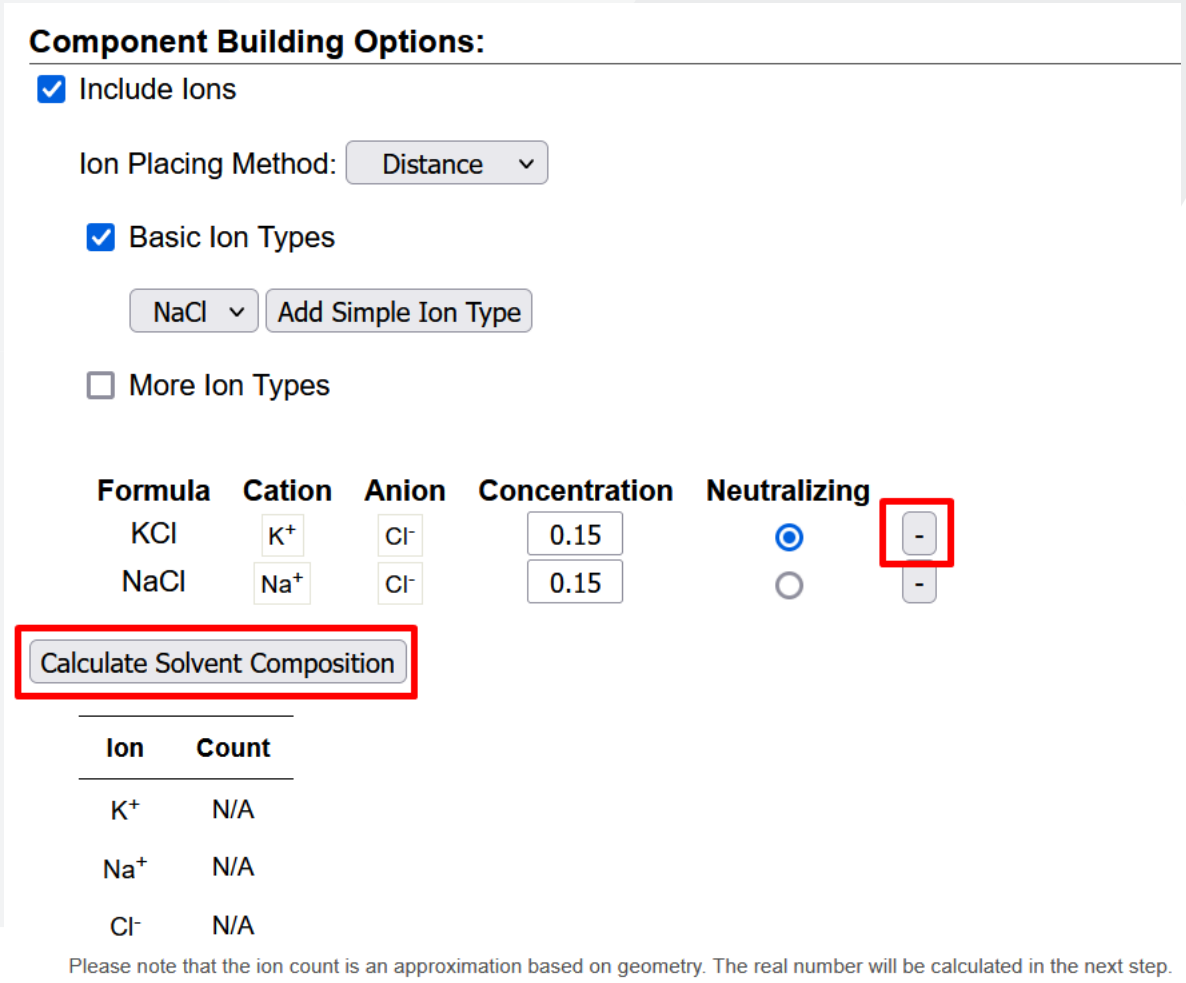
Note: The surface areas listed above are only suggested values. In the case of high cholesterol concentration, 10~20% smaller surface areas will be adopted as default values, which might be more reasonable and close to equilibrium lipid areas. Please refer the manuscript or the following link to list of default surface area we are using.

Next Step:
Determine the System Size

Lehigh University / Department of Biological Sciences / Department of Chemistry / Department of Bioengineering / Im Lab
Problems, Questions, & Comments? / Contact / Forum / Copyright(c) 2006-2024 by the Im Lab



- We will model physiological concentration of salt – 150 mM NaCl



- We will model physiological concentration of salt – 150 mM NaCl

Component Building Options:

Include Ions

Ion Placing Method: Distance

Basic Ion Types

NaCl | Add Simple Ion Type

More Ion Types

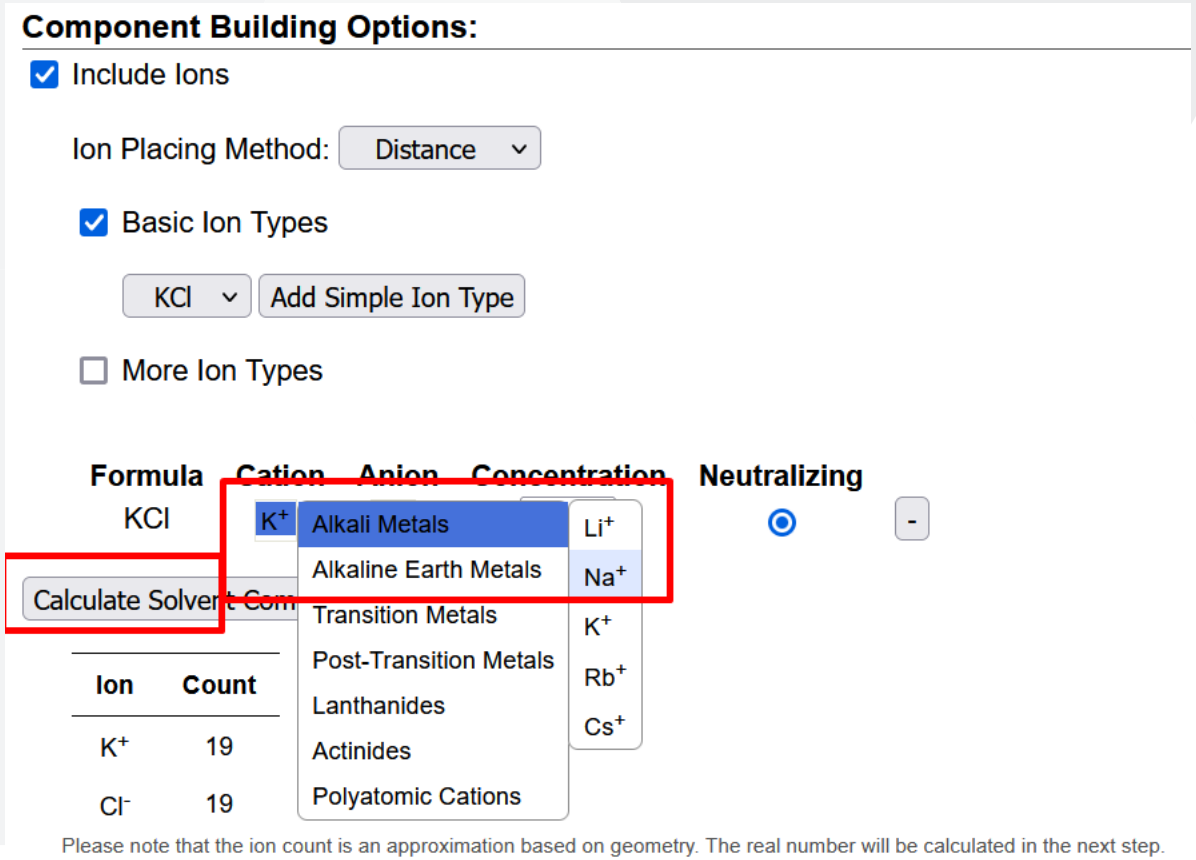
Formula	Cation	Anion	Concentration	Neutralizing	
NaCl	Na^+	Cl^-	0.15	◉	-

Calculate Solvent Composition

Ion	Count
Na^+	19
Cl^-	19

Please note that the ion count is an approximation based on geometry. The real number will be calculated in the next step.

- We will model physiological concentration of salt – 150 mM NaCl



- We will model physiological concentration of salt – 150 mM NaCl

Component Building Options:

☑ Include Ions

Ion Placing Method: Distance

☑ Basic Ion Types

KCl | Add Simple Ion Type

☐ More Ion Types

Formula	Cation	Anion	Concentration	Neutralizing	
NaCl	Na^+	Cl^-	0.15	◉	-

Calculate Solvent Composition

Ion	Count
Na^+	19
Cl^-	19

Please note that the ion count is an approximation based on geometry. The real number will be calculated in the next step.

Next Step:
Build Components

Membrane Builder

User Profile

Bookmark this link, if you want to comeback to this page

STEP 3 | STEP 4 | STEP 5 | STEP 6

JOB ID: 2052637332

download.tgz

Component Input: step4_lipid.inp
Component Output: step4_lipid.out
Component Number: step4_components.str
Component PDB: step4_lipid.pdb (view structure)

Check lipid pentration

The protein surface penetration check finds the lipid tails that go beyond the protein surface, and the lipid ring penetration check detects the lipid tails that pass through the cyclic groups (e.g., cholesterol ring) in the simulation systems. Energy minimization can resolve many of these bad contacts, but one might need to visually check the following lipid molecules to ensure the following contacts are resolved. The user can regenerate the lipid bilayer if necessary.

Lipid ring penetration:

```
No lipid ring penetration is found.
```

Building Ion and Waterbox

Membrane components are generated. Due to time constrains, we first generate the lipid bilayer then generate ions and the water box. Click "Next Step" to generate ions and the water box.

Next Step:
Assemble Components

Membrane Builder

User Profile

Bookmark this link, if you want to comeback to this page

STEP 3 | STEP 4 | STEP 5 | STEP 6

JOB ID: 2052637332

download.tgz

Component Input:	step4_lipid.inp	
Component Output:	step4_lipid.out	
Component Number:	step4_components.str	
Generated Waterbox:	step4.2_waterbox.inp	Input file for water box inclusion
	step4.2_waterbox.out	Output file for water box inclusion
	step4.2_waterbox.crd	CRD file for the water box
Generated Ion:	step4.3_ion.inp	Input file for ion inclusion
	step4.3_ion.out	Output file for ion inclusion
	step4.3_ion.pdb	PDB file for the ion
Component PDB:	step4_lipid.pdb (view structure)	

Assemble Generated Components:

Membrane components are generated. Click "Next Step" to assemble those components together.

Next Step:
Assemble Components

Force Field Options:

CHARMM36m

- [ ] WYF parameter for cation-pi interactions
- [ ] Hydrogen mass repartitioning

  Y. Gao, J. Lee, I.P.S. Smith, H. Lee, S. Kim, Y. Qi, J.B. Klauda, G. Widmalm, S. Khalid, and W. Im *Journal of Chemical Information and Modeling* 2021 61 (2), 831-839. DOI.
- [ ] Multi-site Ca^{2+}

  Currently only available for CHARMM, NAMD, GROMACS, and OpenMM

  Zhang, A., Yu, H., Liu, C. *et al.* The Ca^{2+} permeation mechanism of the ryanodine receptor revealed by a multi-site ion model. *Nat. Commun.* 11, 922 (2020). DOI.

Input Generation Options:

- [x] More CHARMM minimization during input generation
- [ ] NAMD
- [x] GROMACS
- [ ] AMBER
- [ ] OpenMM
- [ ] CHARMM/OpenMM
- [ ] GENESIS
- [ ] Desmond
- [ ] LAMMPS
- [ ] Tinker

Equilibration Options:

- [ ] P21 image transformation (only available for CHARMM)
- [ ] CHARMM DOMDEC (only available for CHARMM)
- (•) Generate grid information for PME FFT automatically
- ( ) Explicit grid information for PME FFT

X Y Z

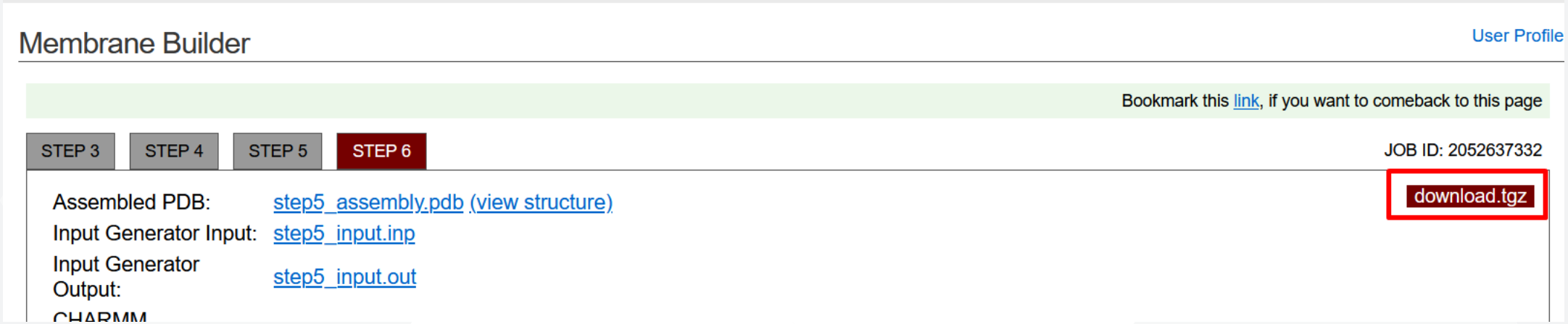
( ) NVT ensemble
(•) NPT ensemble
( ) NPAT ensemble
( ) NPgT ensemble

Surface Tension 0 (dyne/cm)

Temperature: 310.15 K

- Human body temperature

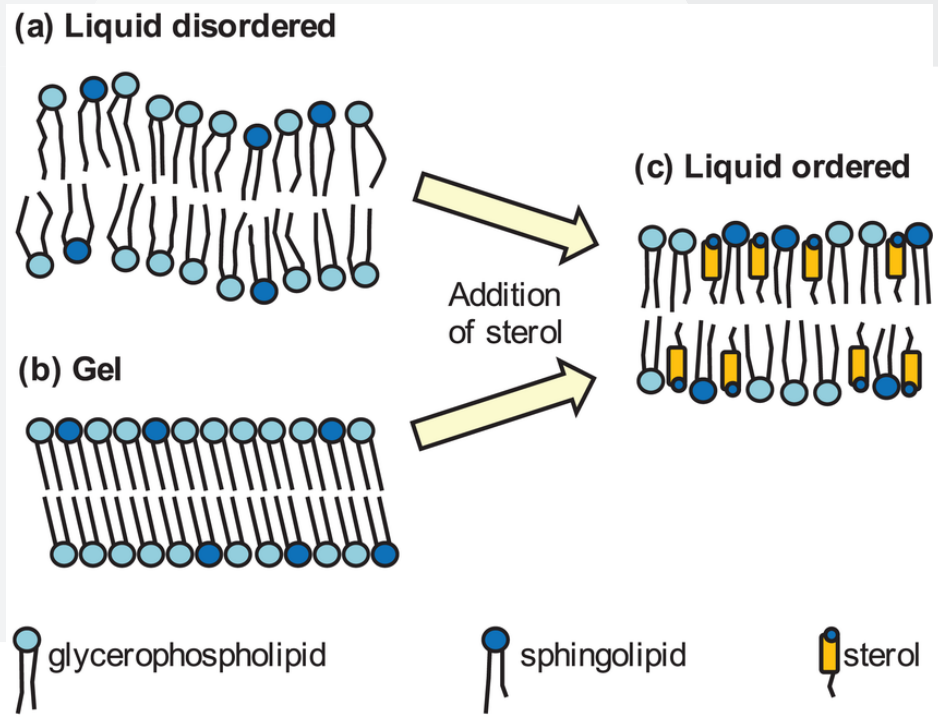
Next Step:
Generate Equilibration and Dynamics Inputs

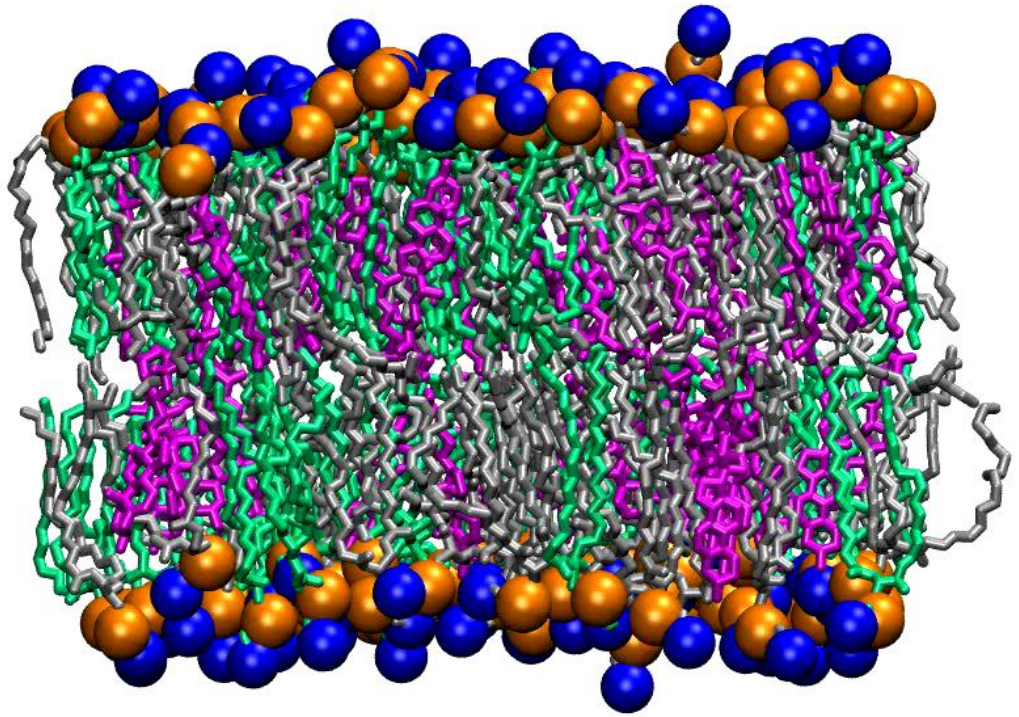

Membrane Builder
User Profile
Bookmark this link, if you want to comeback to this page
STEP 3
STEP 4
STEP 5
STEP 6
JOB ID: 2052637332
Assembled PDB:
step5_assembly.pdb (view structure)
Input Generator Input:
step5_input.inp
Input Generator Output:
step5_input.out
download.tgz

Analysis of MD simulations of lipid membranes

Analysis of MD simulations of lipid membranes

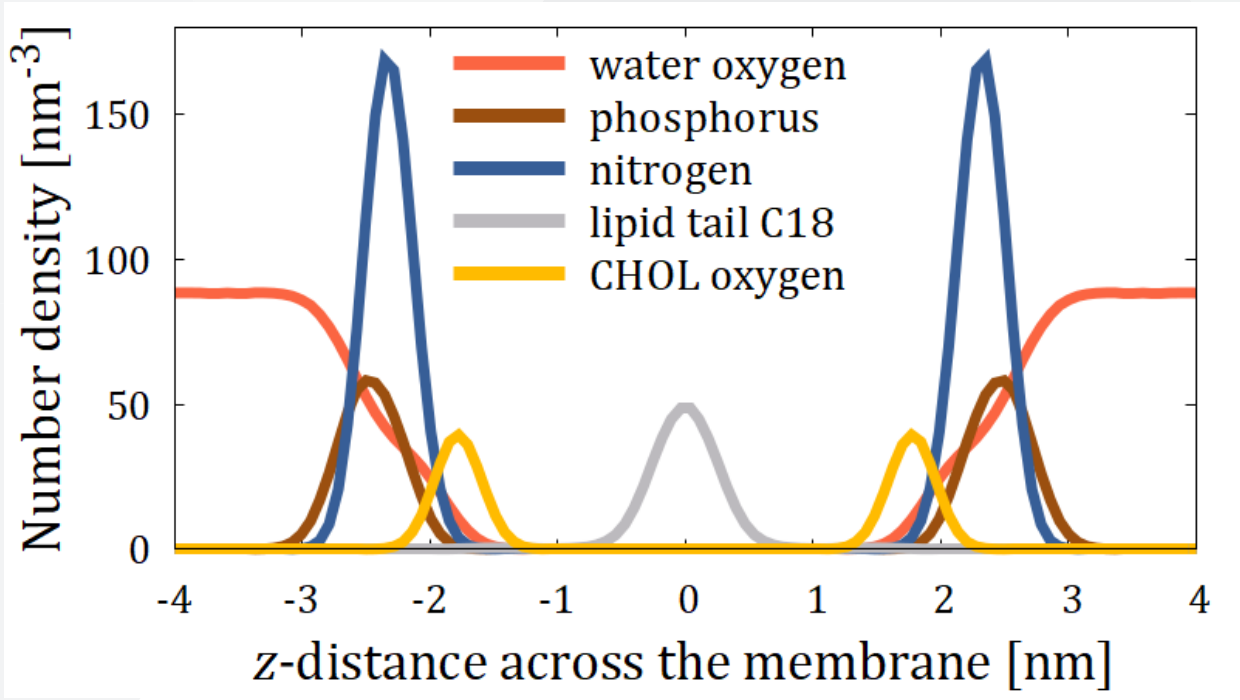
- **POPC; PSM; CHOL**
- **P atoms; N atoms**

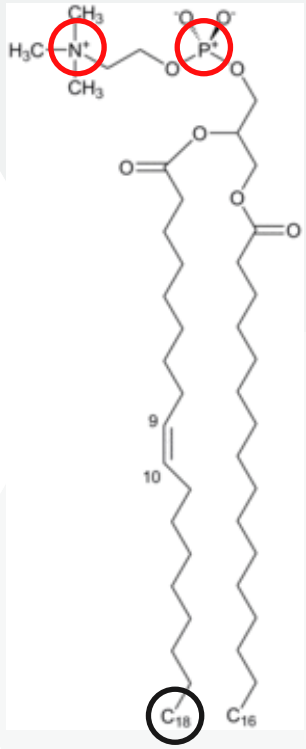




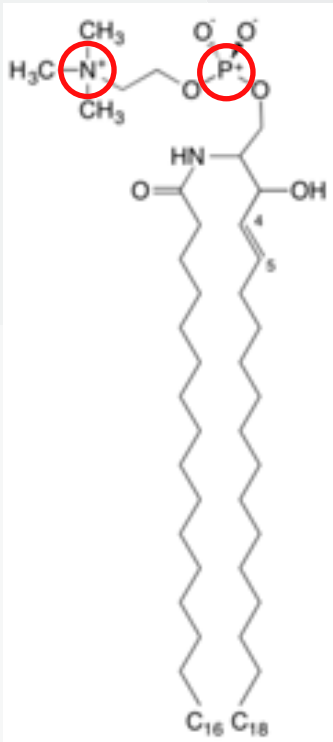
Johnston et al. Yeast 2020, 37 (1), 27–44.

Analysis of MD simulations of lipid membranes

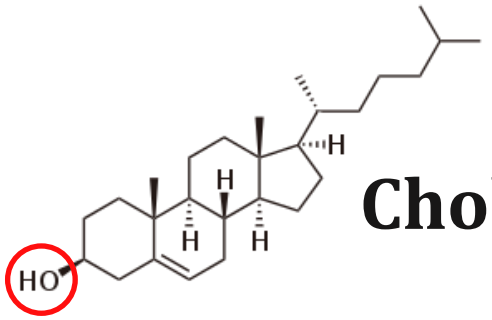




POPC

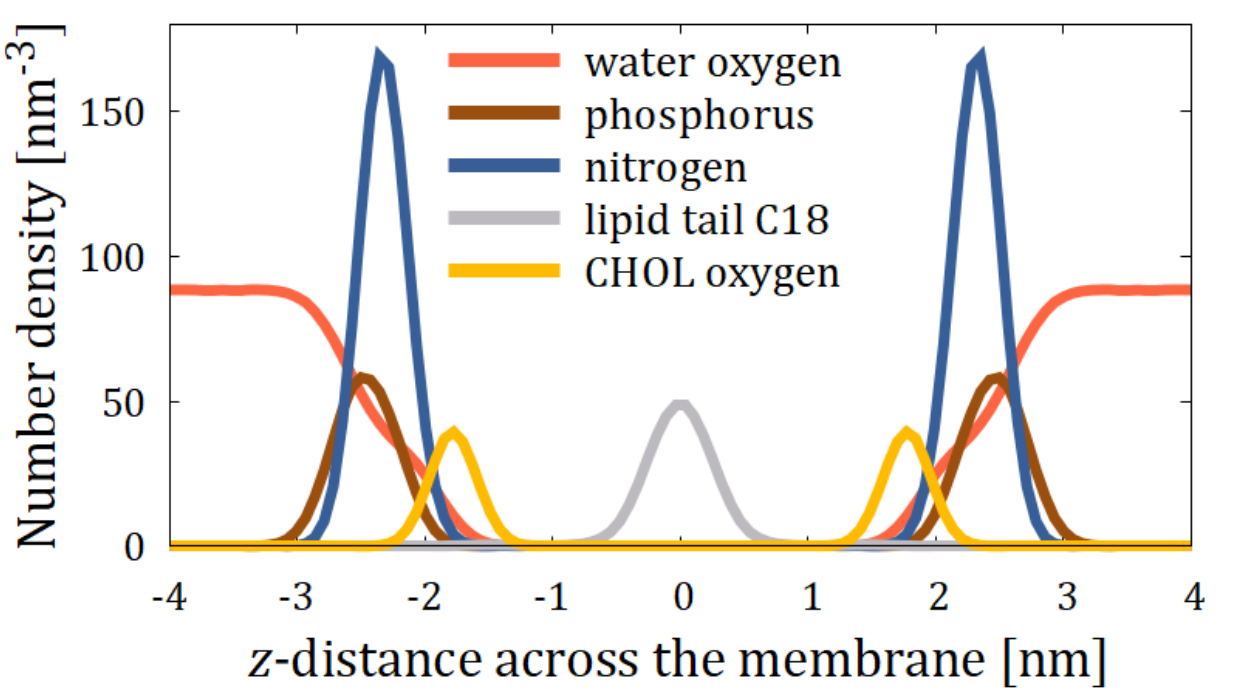


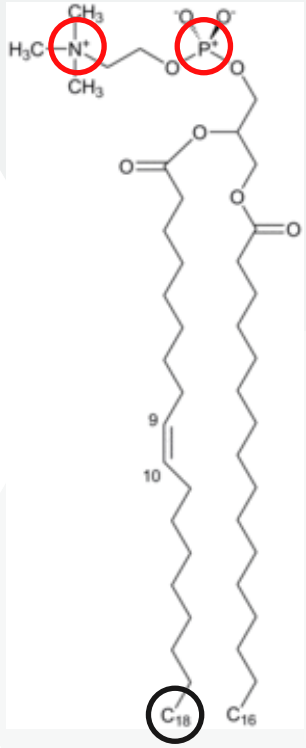
PSM



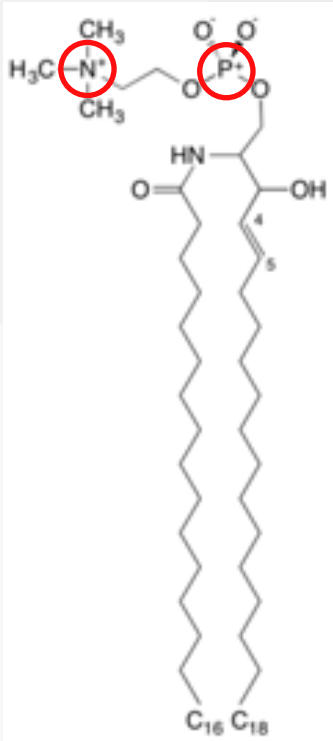
Cholesterol

Analysis of MD simulations of lipid membranes



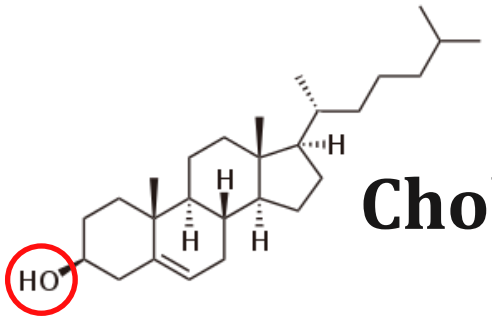


POPC



PSM

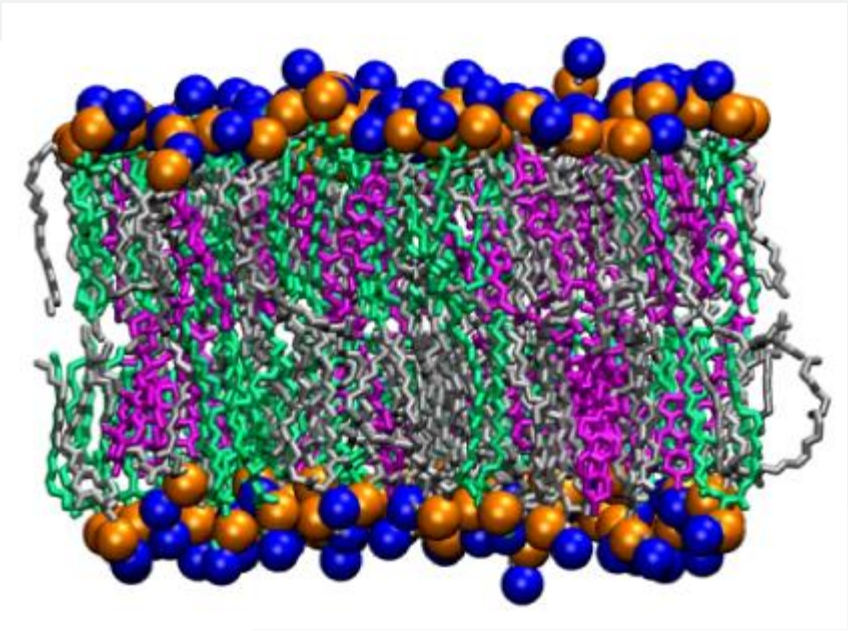
- gmx density -h
- Distribution of atoms along a particular coordinate.
- Often centered and symmetrized for membranes.



Cholesterol

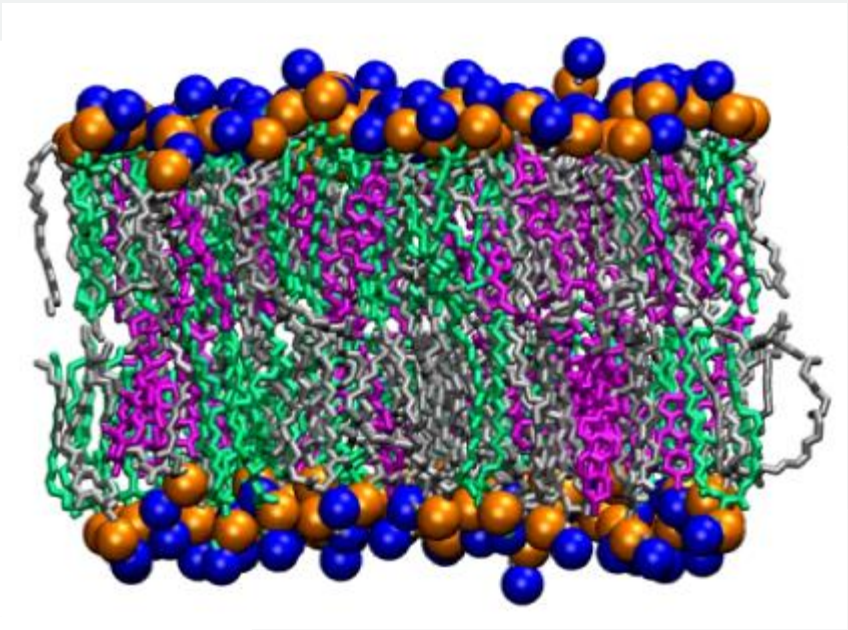
Analysis of MD simulations of lipid membranes

- Area per lipid
  - Typically, box size / number of lipids



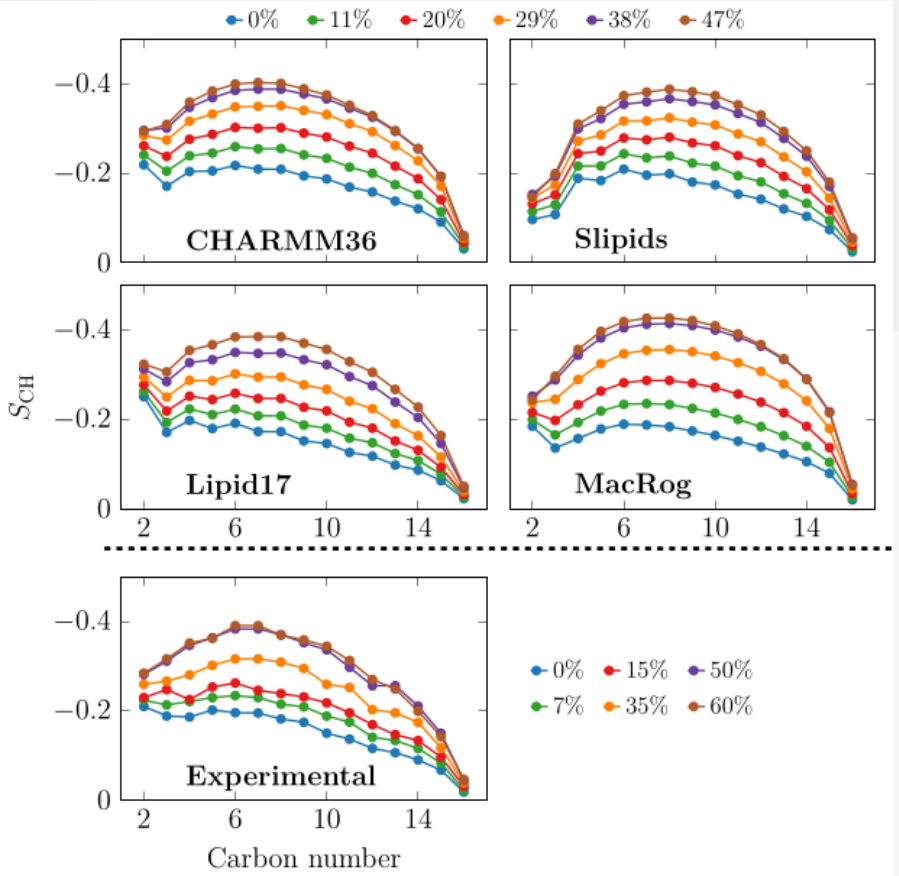
Analysis of MD simulations of lipid membranes

- Area per lipid
  - Typically, box size / number of lipids
- Bilayer thickness
  - Typically, P–P distance



Analysis of MD simulations of lipid membranes

- Area per lipid
  - Typically, box size / number of lipids
- Bilayer thickness
  - Typically, P–P distance
- Order parameter

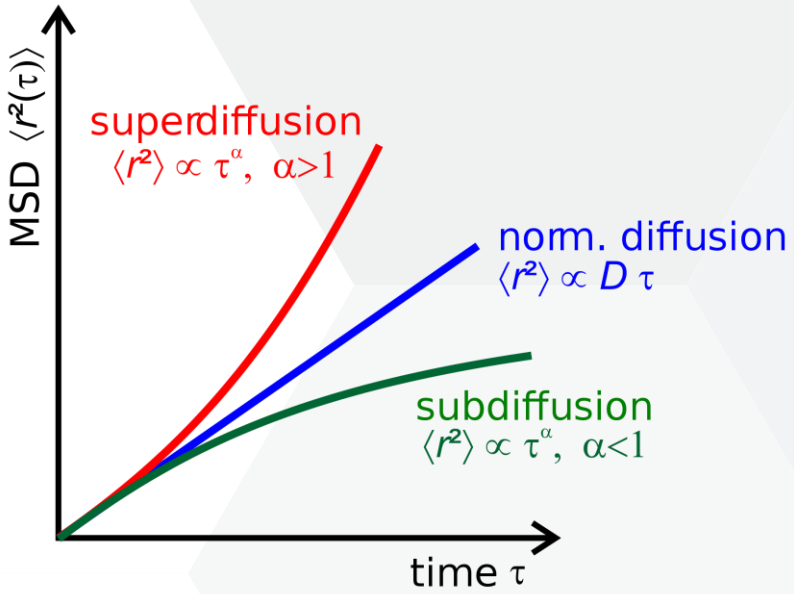


Order parameters for different lipid membranes with varying amount of cholesterol

Javanainen et al. *J. Chem. Theory Comput.* 2023, 19, 6342–6352

Analysis of MD simulations of lipid membranes

- Area per lipid
  - Typically, box size / number of lipids
- Bilayer thickness
  - Typically, P–P distance
- Order parameter
- Diffusion coefficient

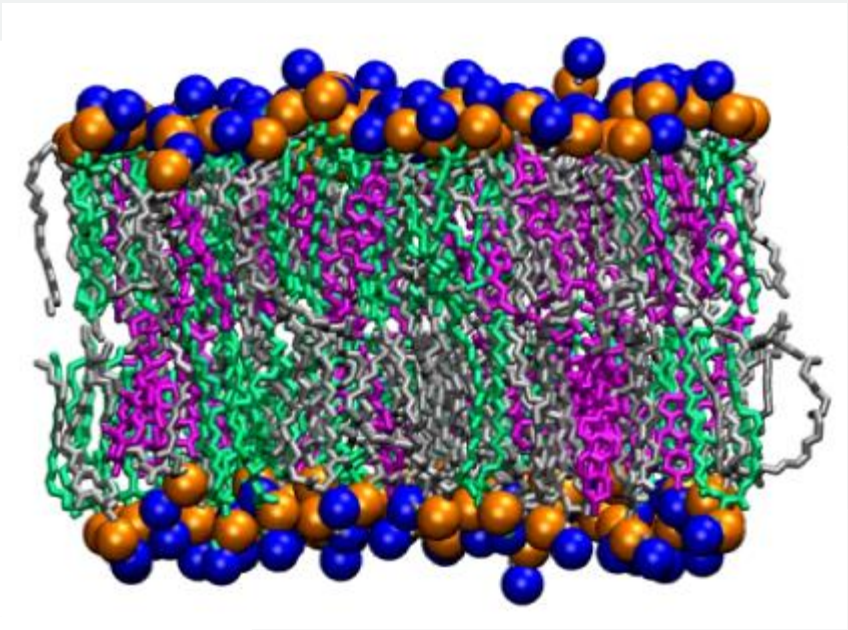


Comparison of normal and anomalous diffusions

taken from Wikipedia

Analysis of MD simulations of lipid membranes

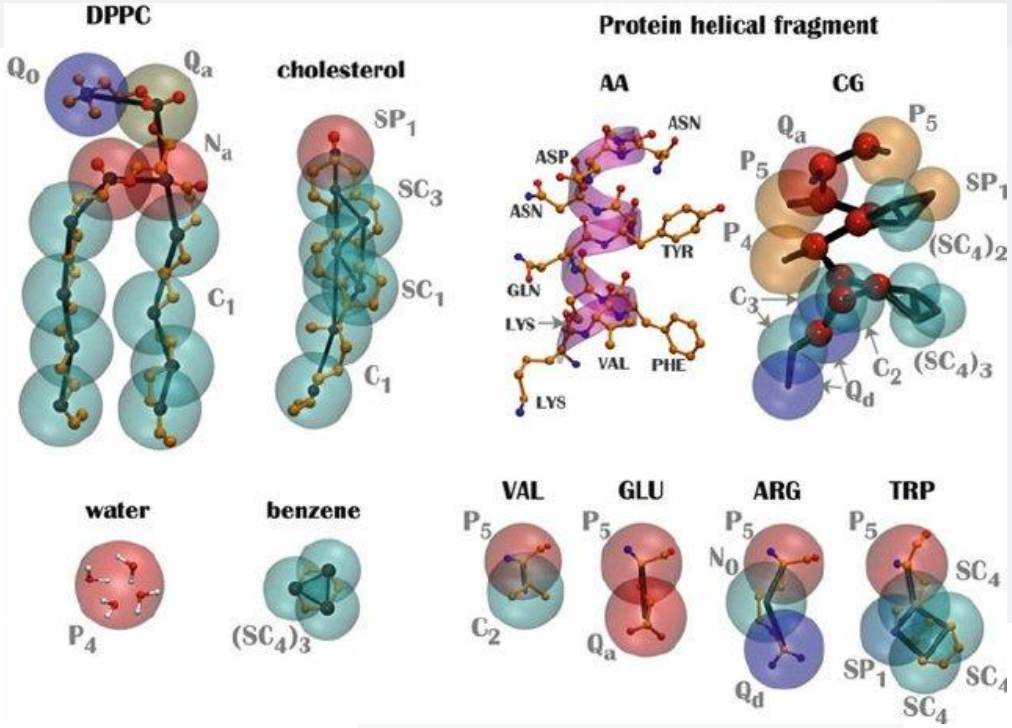
- Area per lipid
  - Typically, box size / number of lipids
- Bilayer thickness
  - Typically, P–P distance
- Order parameter
- Diffusion coefficient
- Bending modulus and elastic properties
- Surface tension
- Tilt angle



Martini force field: principles and applications

Martini force field: principles and applications

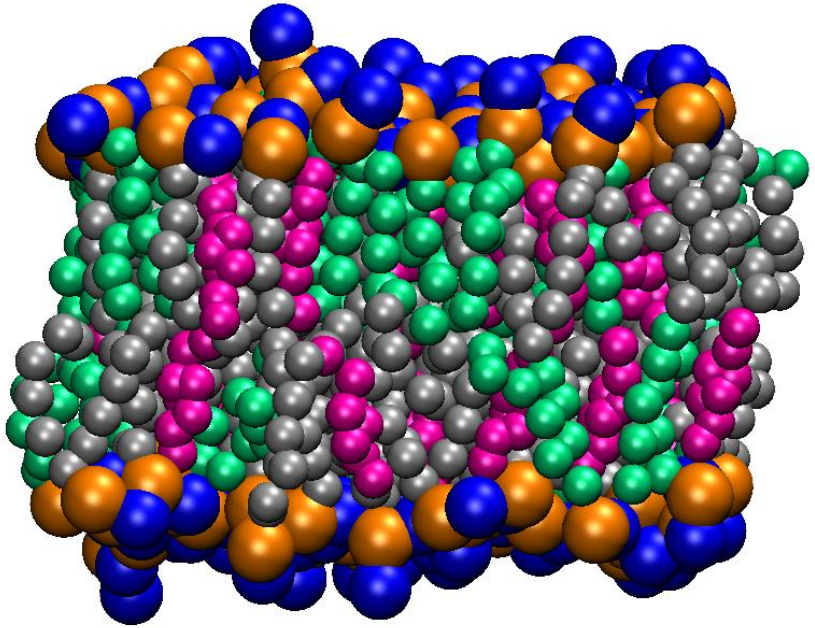
- Martini is a coarse-grained force field. Coarse-graining is grouping multiple atoms into single “beads”.

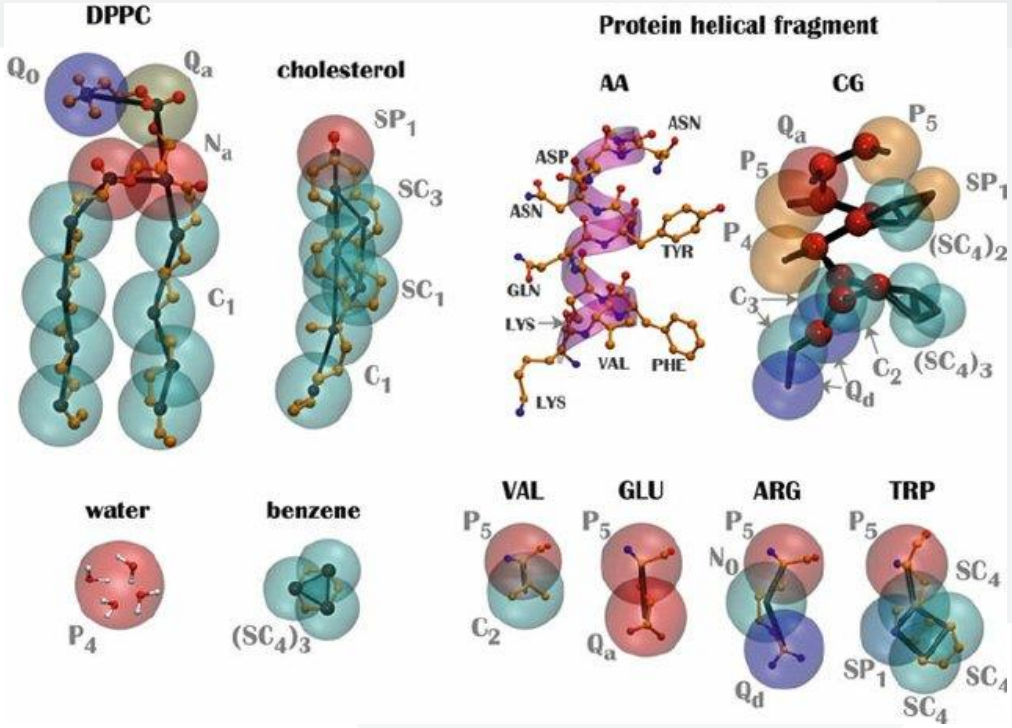


[Periole & Marrink, Methods in Molecular Biology, 924:533-65]

Martini force field: principles and applications

- Martini is a coarse-grained force field. Coarse-graining is grouping multiple atoms into single “beads”. Initially was created for lipids.

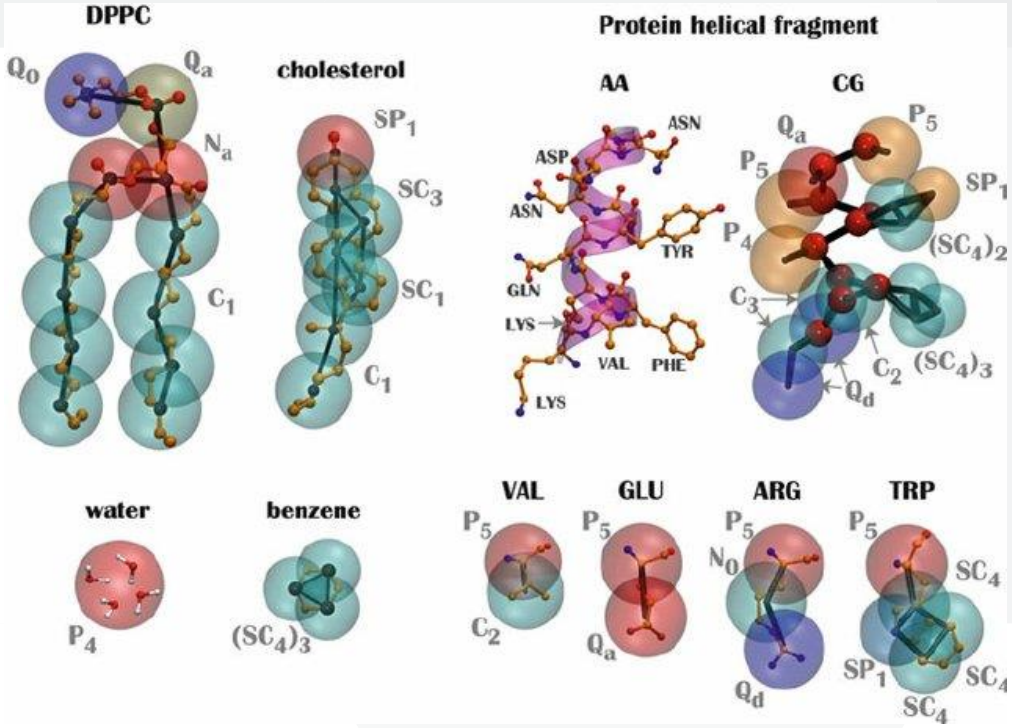




[Periole & Marrink, Methods in Molecular Biology, 924:533-65]

Martini force field: principles and applications

- Martini is a coarse-grained force field. Coarse-graining is grouping multiple atoms into single “beads”.
- Allows larger systems and longer timescales compared to all-atom simulations.

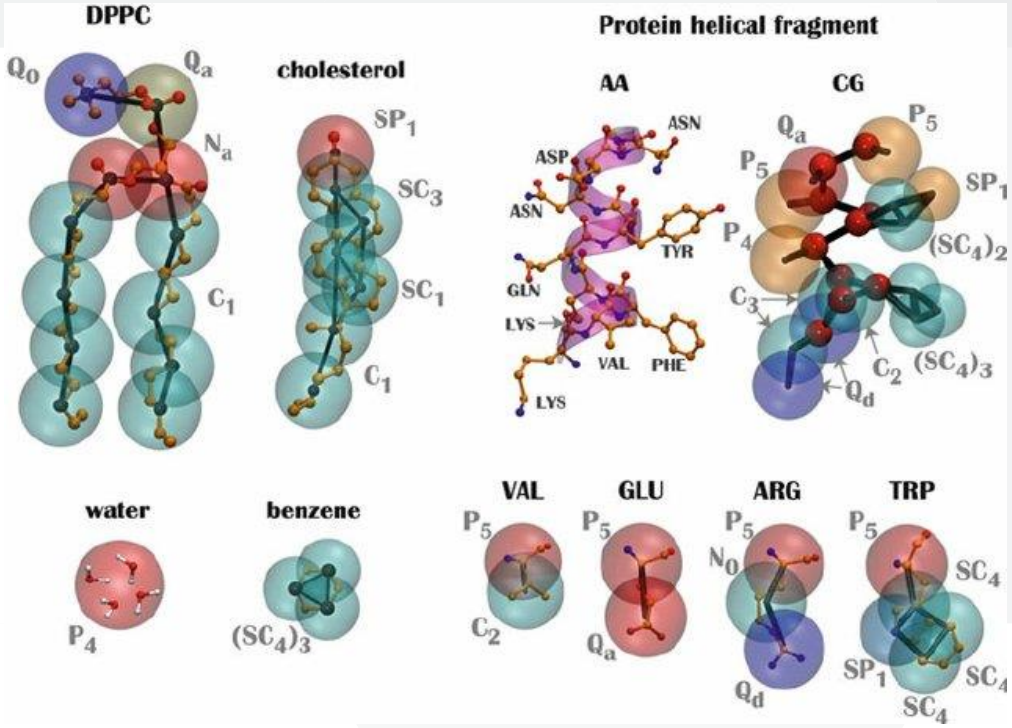


[Periole & Marrink, Methods in Molecular Biology, 924:533-65]

Martini force field: principles and applications

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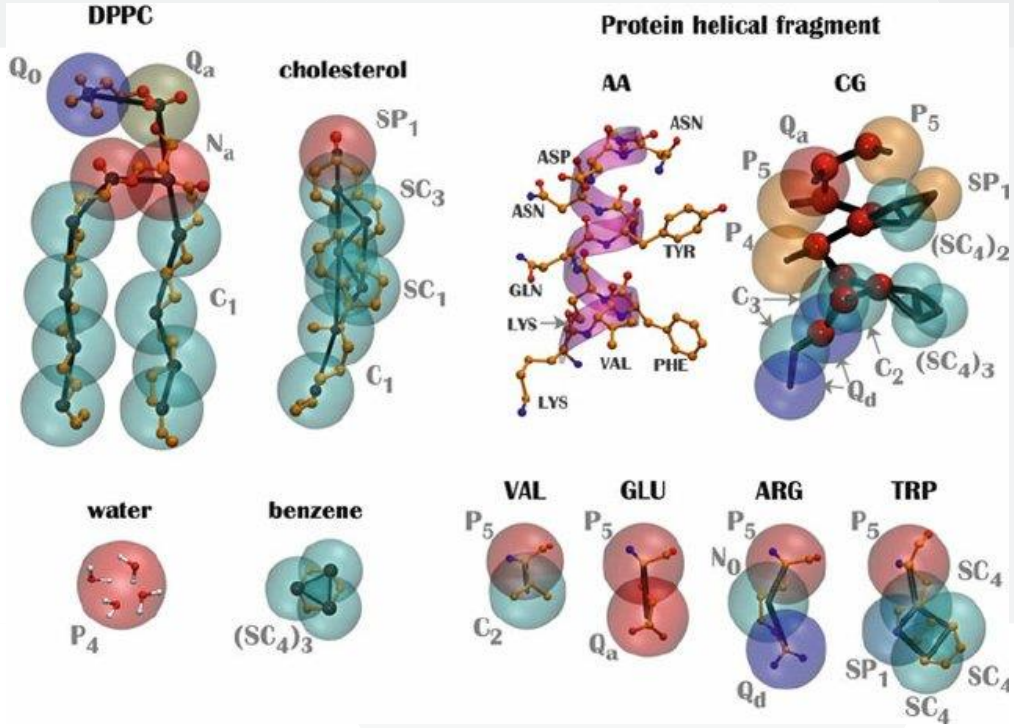
- Allows larger systems and longer timescales compared to all-atom simulations.
  - Our membrane system on an advanced machine equipped with GPU.
  - All-atom – **200 ns/day**.
  - Coarse-grained – **20000 ns/day**.
  - Two orders of magnitude difference.



[Periole & Marrink, Methods in Molecular Biology, 924:533-65]

Martini force field: principles and applications

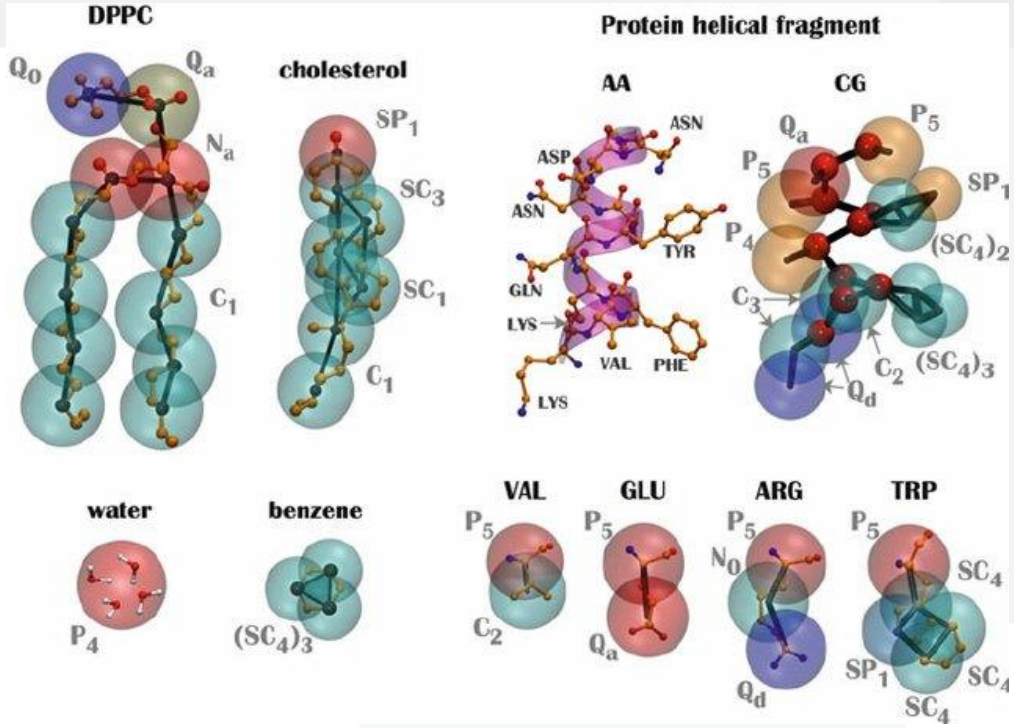
- Martini is a coarse-grained force field. Coarse-graining is grouping multiple atoms into single “beads”.
- Allows larger systems and longer timescales compared to all-atom simulations.
- Martini is highly versatile: lipids, proteins, nucleic acids, carbohydrates, small molecules, nanomaterials.



[Periole & Marrink, Methods in Molecular Biology, 924:533-65]

Martini force field: principles and applications

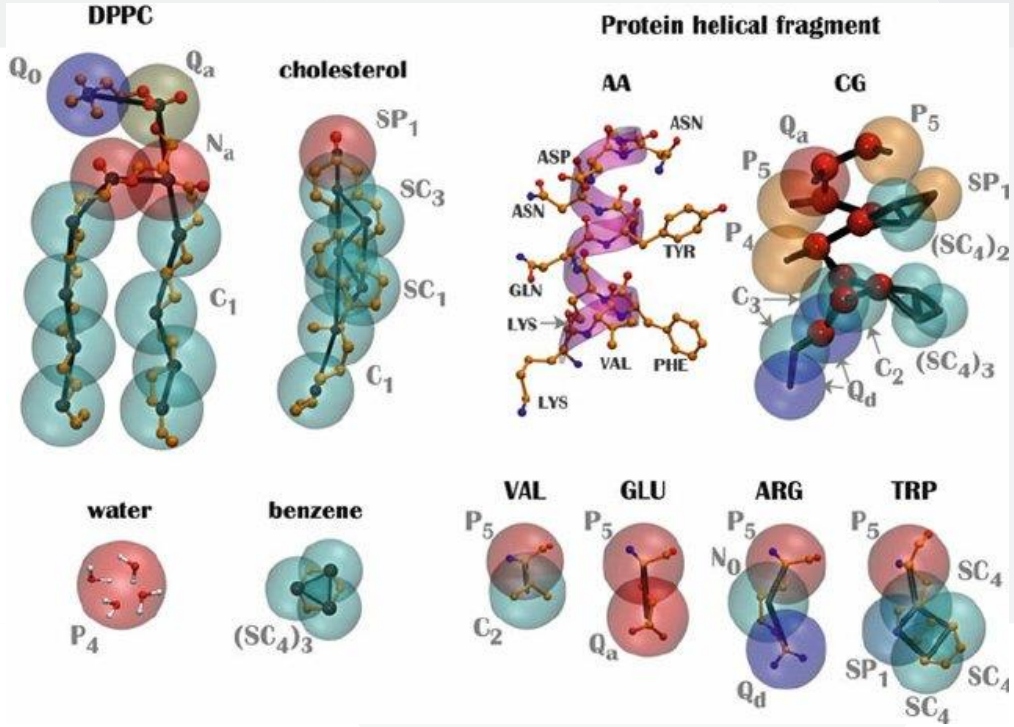
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- Allows larger systems and longer timescales compared to all-atom simulations.
- Martini is highly versatile: lipids, proteins, nucleic acids, carbohydrates, small molecules, nanomaterials.
- Martini uses a specific parameterization that maps atomistic structures to coarse-grained representations.



[Periole & Marrink, Methods in Molecular Biology, 924:533-65]

Martini force field: principles and applications

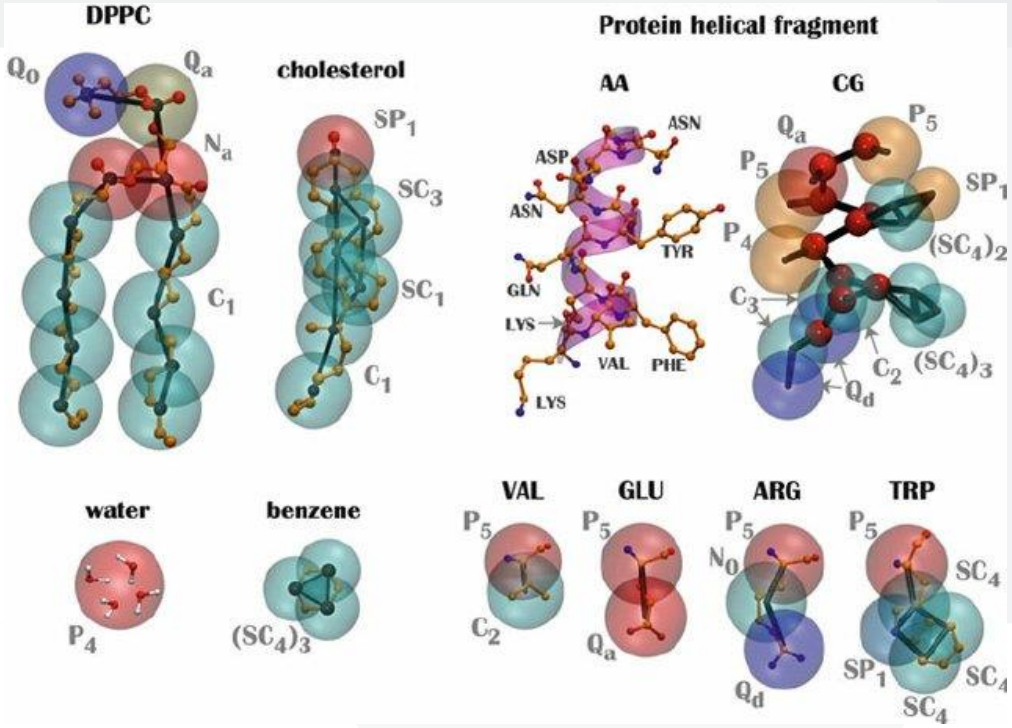
- Martini is a coarse-grained force field. Coarse-graining is grouping multiple atoms into single “beads”.
- Allows larger systems and longer timescales compared to all-atom simulations.
- Martini is highly versatile: lipids, proteins, nucleic acids, carbohydrates, small molecules, nanomaterials.
- Martini uses a specific parameterization that maps atomistic structures to coarse-grained representations.
- Involves calibrating interaction potentials to reproduce key properties.



[Periole & Marrink, Methods in Molecular Biology, 924:533-65]

Martini force field: principles and applications

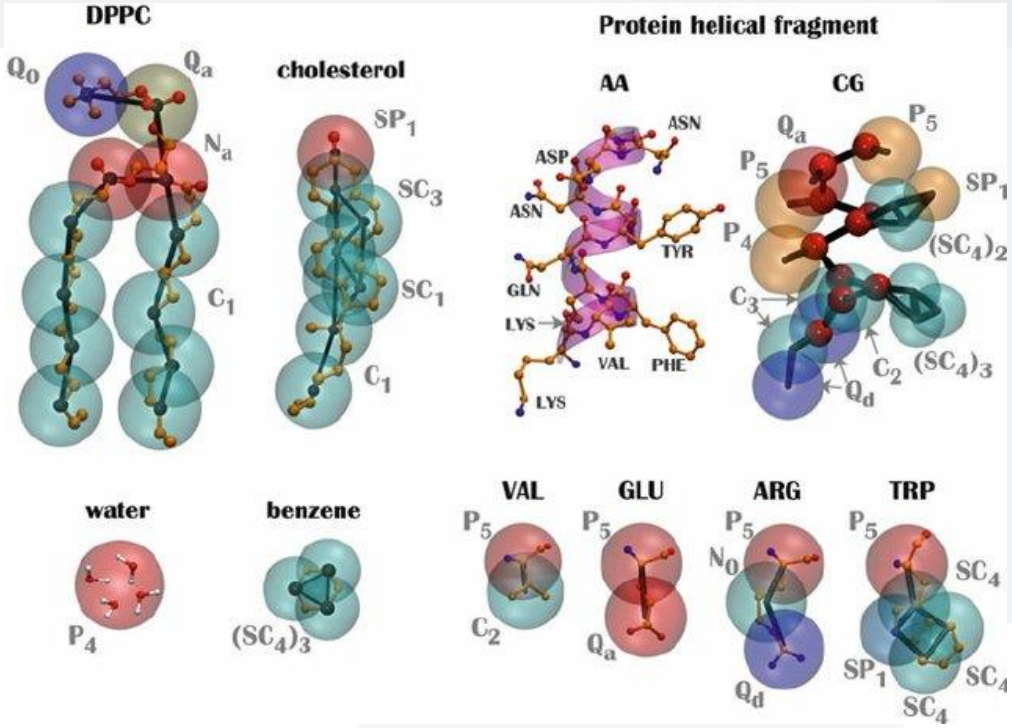
- By reducing the degrees of freedom and smoothing out the energy landscape, Martini enhances the efficiency of sampling conformational space.



[Periole & Marrink, Methods in Molecular Biology, 924:533-65]

Martini force field: principles and applications

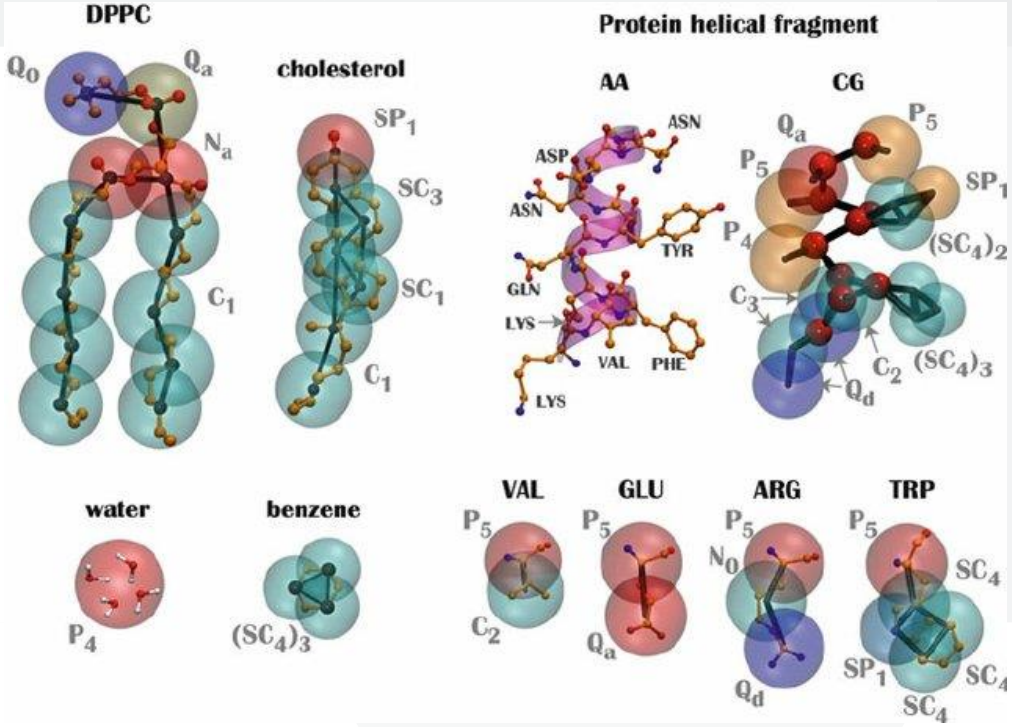
- By reducing the degrees of freedom and smoothing out the energy landscape, Martini enhances the efficiency of sampling conformational space.
- It allows to study membrane fusion, self-assembly, and other processes that are challenging for all-atom models.



[Periole & Marrink, Methods in Molecular Biology, 924:533-65]

Martini force field: principles and applications

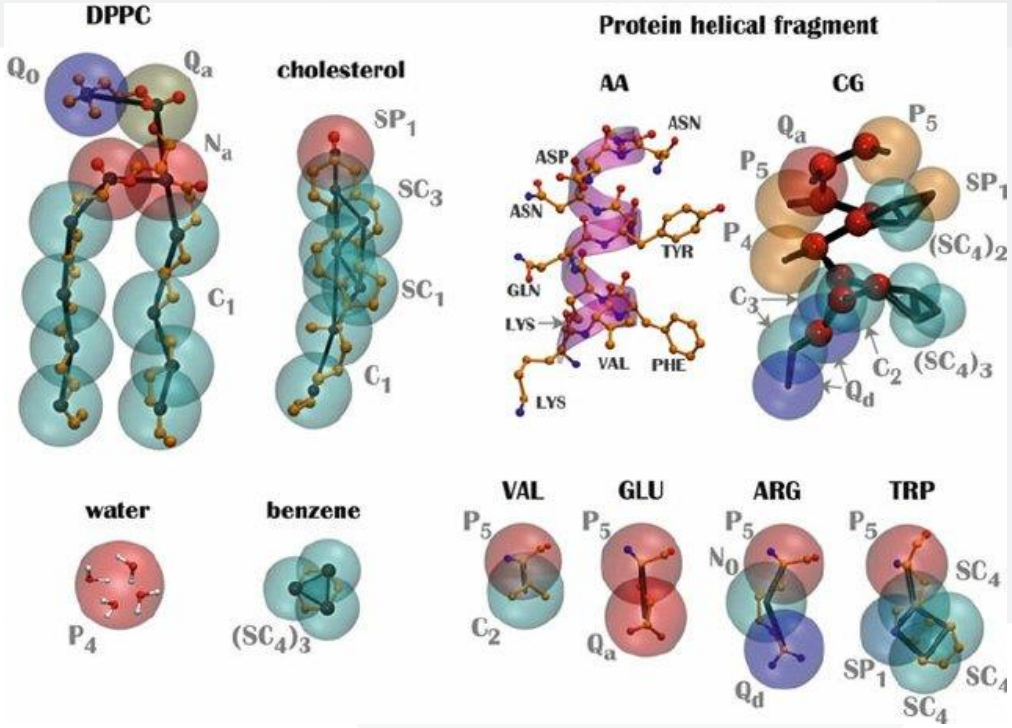
- By reducing the degrees of freedom and smoothing out the energy landscape, Martini enhances the efficiency of sampling conformational space.
- It allows to study membrane fusion, self-assembly, and other processes that are challenging for all-atom models.
- Martini force field is supported by an active research community that continuously updates and refines the parameters and methodologies.



[Periole & Marrink, Methods in Molecular Biology, 924:533-65]

Martini force field: principles and applications

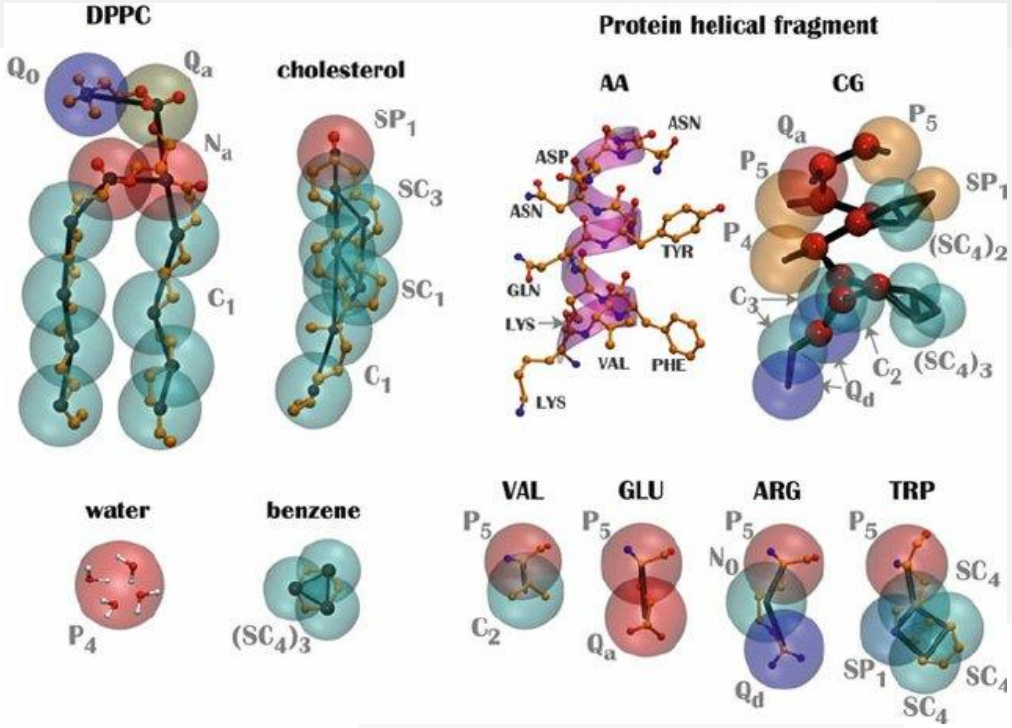
- Martini has drawbacks!



[Periole & Marrink, Methods in Molecular Biology, 924:533-65]

Martini force field: principles and applications

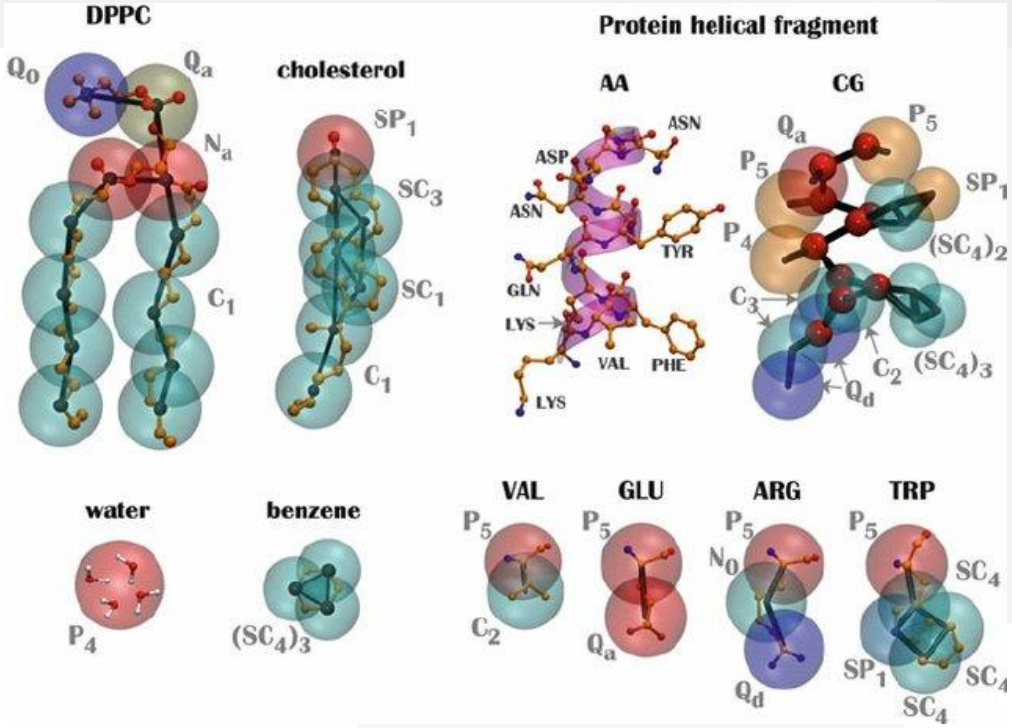
- Martini has drawbacks!
- There are systems that cannot be studied (or challenging to study) by all-atom simulations.
  - Processes involving charge transfer.
  - Chemical reactions.



[Periole & Marrink, Methods in Molecular Biology, 924:533-65]

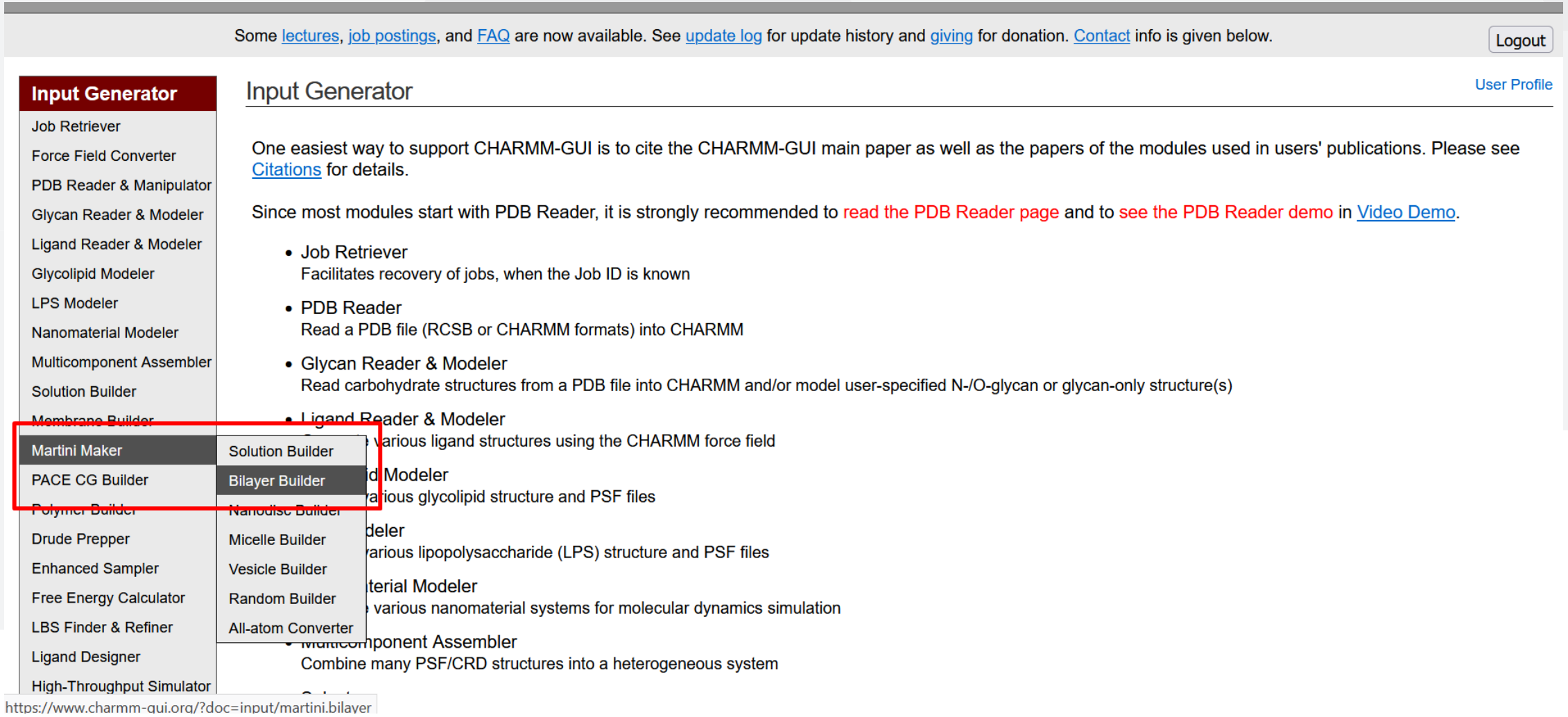
Martini force field: principles and applications

- Martini has drawbacks!
- There are systems that cannot be studied (or challenging to study) by all-atom simulations.
  - Processes involving charge transfer.
  - Chemical reactions.
- There are even such systems for coarse-grained simulations including Martini.
  - Secondary structure of proteins.
  - Polar and charged interactions.
  - Temperature-related processes.
  - Basically everything, where chemical specificity is essential.
  - Certain lipid membrane processes.

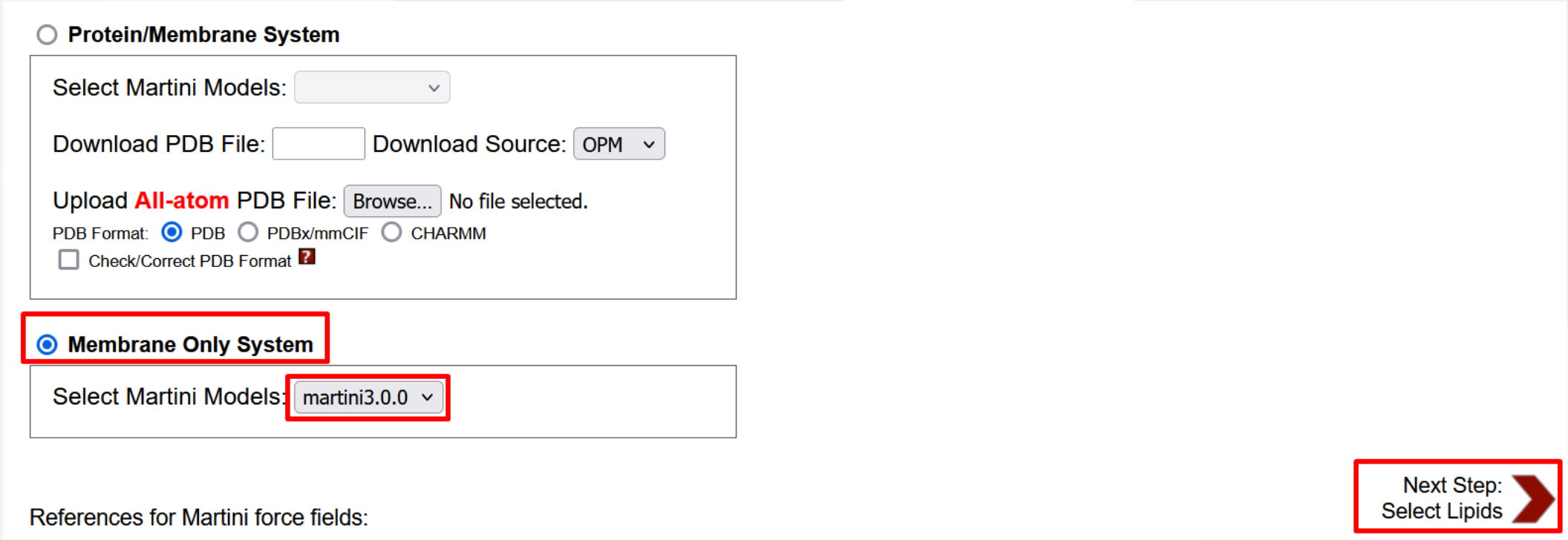


[Periole & Marrink, Methods in Molecular Biology, 924:533-65]

Martini in CHARMM-GUI



Martini in CHARMM-GUI



Martini tools: martinize

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Computational Lipidomics with *insane*: A Versatile Tool for Generating Custom Membranes for Molecular Simulations

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insane

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insane.py

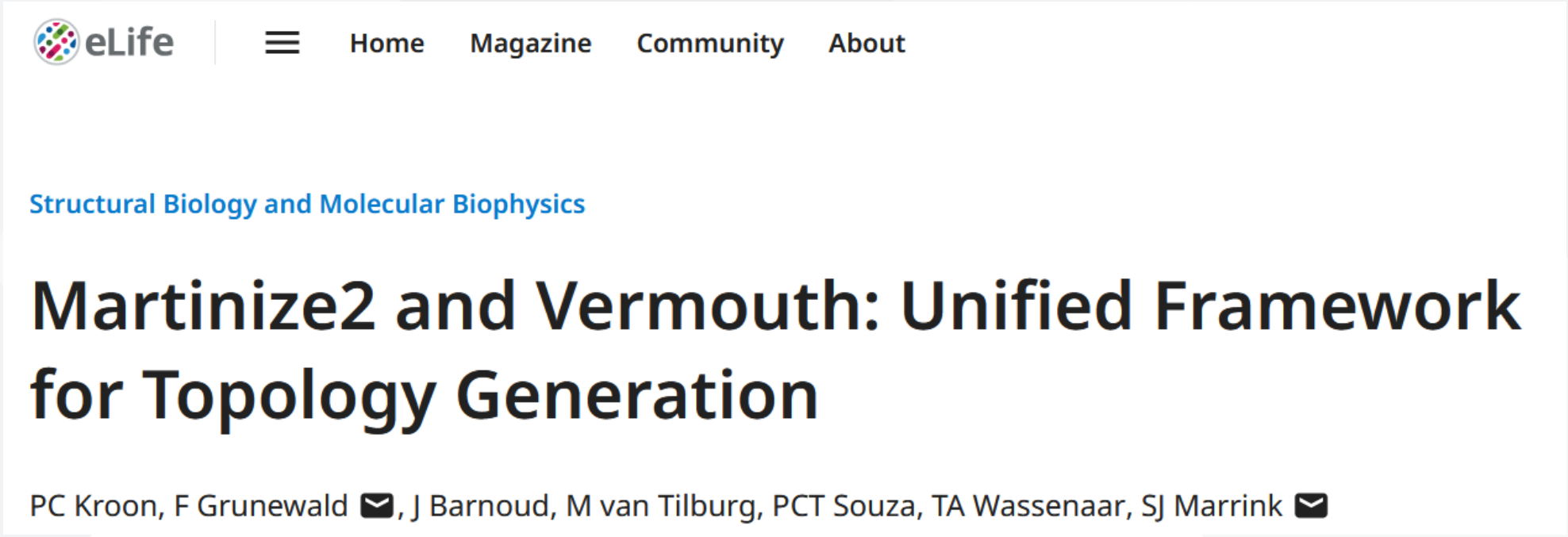
insane.py is a python script to setup Martini bilayer systems. It handles a broad range of different lipid types, is very flexible with respect to box types and can insert a (CG) protein structure in the bilayer if one is given. Help text and options are obtained by typing: insane.py -h

Read much more about *insane* from the *insane* paper: DOI: 10.1021/acs.jctc.5b00209

2015.06.25 – *insane* was updated adding a few features, fixing a few bug and the lipid templates now use the new lipid-naming schema, the old templates are still available with a .o after the name e.g. “DHPC.o”.

- Creates lipid membranes and also proteins inside them

Martini tools: martinize



- Converts all-atom structures into Martini
- Generates inputs for GROMACS since Martini is GROMACS-oriented (at least, for now)
- Martinize2 generates topologies for both Martini 2 and Martini 3 models